
HydDown CO2 Release — Technical Reference

Release 2.0

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Physical models, thermodynamics, heat and mass transfer, and validation
of the CO₂ release / dry-ice add-on to HydDown.

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This technical reference documents the CO₂ **release and dry-ice** add-on developed on the `co2-release-hem` branch of HydDown [1, 2]. It covers the additional physical models, the all-CoolProp thermodynamic backend with its precomputed solid-CO₂ property table, the homogeneous-equilibrium (HEM) and non-equilibrium (HNE) discharge models, the in-vessel and atmospheric dry-ice estimation, the heat- and mass-transfer treatment, and the validation against the CARDICE (Ineris 2 m³ sphere) and the Høydalsvik/Munkejord (SINTEF dense-phase) experiments.

It is a companion to the base HydDown *Manual* (`Manual.md`); only the CO₂ extension is described here.

INTRODUCTION

1.1 Motivation

In carbon capture and storage (CCS) with ship or truck transport, CO₂ is handled as a pressurised, liquefied fluid at vapour/liquid equilibrium. An accidental leak - at an onshore tank, on a ship, or during transfer - depressurises the inventory towards atmospheric pressure. Because the atmospheric pressure of CO₂ (1 atm) lies **below the triple point** ($P_{tp} = 5.18$ bar, $T_{tp} = -56.6$ °C), the expanding stream necessarily passes into the **vapour + solid** region: it forms **dry ice** (“CO₂ snow”) both downstream of the leak and, for a vapour-space release, as a large retained bank inside the vessel. Dry-ice formation drives long shutdowns (sublimation is slow) and severe low wall temperatures that govern material selection.

The base HydDown code [1, 2] uses CoolProp [3] as its thermodynamic backend. CoolProp’s reference CO₂ equation of state [4] has **no solid phase** and is guarded at the triple point, so out of the box it can neither integrate a homogeneous-equilibrium (HEM) discharge isentrope through a sub-triple throat nor evaluate the atmospheric dry-ice end state. The CO₂ release add-on removes both limitations **without a second runtime backend**: CoolProp is made to extrapolate its fluid EoS into the metastable gas region below the triple point (imposed-phase evaluation), and a small precomputed solid-CO₂ property table supplies the one thing CoolProp lacks - the solid branch. Below the triple point pure CO₂ has a single degree of freedom, so every “flash” reduces to lever-rule arithmetic on the sublimation line (*Thermodynamics and the Solid Lever*).

1.2 Scope

The add-on is tailored to pure CO₂ and to two release geometries:

- **Gas (vapour-space) release** - a hole in the vapour space; saturated-vapour stagnation. The liquid stays in the vessel, cools to the triple point, freezes to a large retained dry-ice bank, and the vessel finally blows down on the sublimation line. This is the classic “hundreds of kilograms of snow in the vessel” case.
- **Liquid (bottom) release** - a hole in the liquid space; saturated- or dense-liquid stagnation. The liquid drains (and boils off) and is essentially gone *before* the vessel reaches the triple point, so **no significant in-vessel dry ice** forms; the dry ice appears downstream of the orifice instead.

Both cases require the two-phase (non-equilibrium) vessel model (`calculation.non_equilibrium: true`, `calculation.type: energybalance`), a liquid fill (`vessel.liquid_level`), and the new top-level `release: block`. The add-on is validated against two independent experimental campaigns - the low-pressure CARDICE sphere and the high-pressure Høydalsvik/Munkejord dense-phase vessel (*Experimental Basis*).

1.3 What the add-on provides

Table 1.1: Physical models added on co2-release-hem

Area	Model
Thermodynamics	all-CoolProp fluid (native above the triple point; imposed-phase extrapolation below it) plus a precomputed solid-CO ₂ property table; a self-consistent triple point and sublimation line; lever-rule flashes (<i>Thermodynamics and the Solid Lever</i>).
Discharge	homogeneous-equilibrium (HEM) rate with a solid-aware throat; an optional non-equilibrium (HNE) liquid boost N for delayed flashing, with a pressure-scaled fade; phase-split discharge coefficients (<i>Discharge Model</i>).
Dry ice	atmospheric (downstream) vapour/solid split by an isenthalpic lever; an opt-in in-vessel solid model (two-zone triple-point plateau and sublimation descent, with a mass-derived gas/solid interface) (<i>Dry-Ice Estimation</i>).
Heat transfer	a two-node (gas-contact / condensate) wall; Churchill-Chu free convection for the below-triple gas wall; Cooper/Rohsenow boiling on the wetted wall during the plateau; a sublimation-tracking dry-ice wall on the descent (<i>Heat and Mass Transfer</i>).
Mass / energy	separate gas- and liquid-zone balances with phase-attributed outflow enthalpy for a liquid- vs vapour-space release (<i>Heat and Mass Transfer</i>).
Validation	the full CARDICE pure-CO ₂ set (tests 5-10, 10-20 bar) and the Høy-dalsvik/Munkejord dense-phase set (120 bar, riser and no-riser) (<i>Validation</i>).

1.4 Architecture

The CO₂ physics lives in a standalone module, `src/hyddown/co2_release.py` (the lever/table/flash mathematics, class `C02ReleaseModel`) with a **CoolProp-backed implementation** `co2_release_cp.py` (`C02ReleaseModelCP`) that supplies every thermodynamic call from CoolProp plus the solid table `hyddown.co2_solid`. The runtime therefore needs **only CoolProp and numpy**. The main HydDown class in `hdclass.py` calls thin glue, so the CO₂ machinery is never imported for a non-release run. The vessel state above the triple point is tracked by the existing CoolProp non-equilibrium model (NEM); once the tank approaches the triple point the state is handed to the below-triple two-zone model.

Note

The solid-CO₂ property table (`hyddown.co2_solid`) is generated **offline, once**, by `scripts/gen_co2_solid.py` from a solid-CO₂ model in thermopack [5] (GERG-2008 fluid basis), then re-referenced into CoolProp's enthalpy/entropy basis via the physical latent heat of sublimation. The table is committed to the repository; thermopack is **not** a runtime dependency.

A minimal input looks like:

```
vessel:    {length: 0.0, diameter: 1.58, type: Hemispherical,
            liquid_level: 0.62, thickness: 0.072, ...}
initial:   {temperature: 253.15, pressure: 2000000, fluid: "CO2",
            gas_temperature: 256.2}                # superheated gas zone
→(optional)
```

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```

calculation: {type: energybalance, non_equilibrium: true,
              time_step: 20, end_time: 60000, h_gas_liquid: calc_two_sided}
valve:       {type: none, flow: discharge, back_pressure: 101325.}
release:     # <-- the CO2 add-on
              # or "gas"
  type: liquid
  diameter: 0.003
  discharge_coef: 0.62 # liquid (flashing) orifice
  discharge_coef_gas: 0.84 # gas tail
  liquid_nonequilibrium: 0.174 # HNE boost (liquid only)
  eos: CoolProp
  solid_in_vessel: true
  solid_h_gas_wall: churchill # below-triple gas-wall
  ↪ convection
  solid_h_inner: cooper # plateau boiling on the
  ↪ wetted wall
  solid_h_gas_solid: 3.0 # gas<->dry-ice interphase HTC
  back_pressure: 101325.
  atm_pressure: 101325.
heat_transfer: {type: specified_h, temp_ambient: 283.15,
                h_outer: 0.4, h_inner: calc}

```

Worked validation cases ship in validation/reconciled/CARDICE_*.yaml (Ineris) and validation/munkejord/*.yaml (SINTEF); the figure generators are scripts/validate_cardice_reconciled.py and scripts/validate_munkejord.py. A full list of release: keys and their defaults is in *Nomenclature and Input Schema*.

MODEL MAP (PHASE DECISION TREE)

A CO₂ blowdown moves through a sequence of thermodynamic regimes, and the add-on switches the **discharge model, vessel (non-equilibrium) treatment, heat-transfer law and phase lever** as it crosses each regime boundary. This chapter is the road map: the decision tree below shows the phases and their transitions, and the table that follows lists which model is in force in each phase. The individual models are derived in *Thermodynamics and the Solid Lever* (backend), *Discharge Model* (mass flux), *Dry-Ice Estimation* (phase lever) and *Heat and Mass Transfer* (wall / interphase).

The starting state can be **dense-phase / sub-cooled liquid** (e.g. the 120 bar Høydalsvik/Munkejord tests) or **saturated two-phase** liquid + vapour (e.g. the 10-20 bar CARDICE tests); both enter the same tree. The first branch is the **release location** - a vapour-space hole (gas draw) or a liquid-space hole (liquid draw) - which selects the left or right spine.

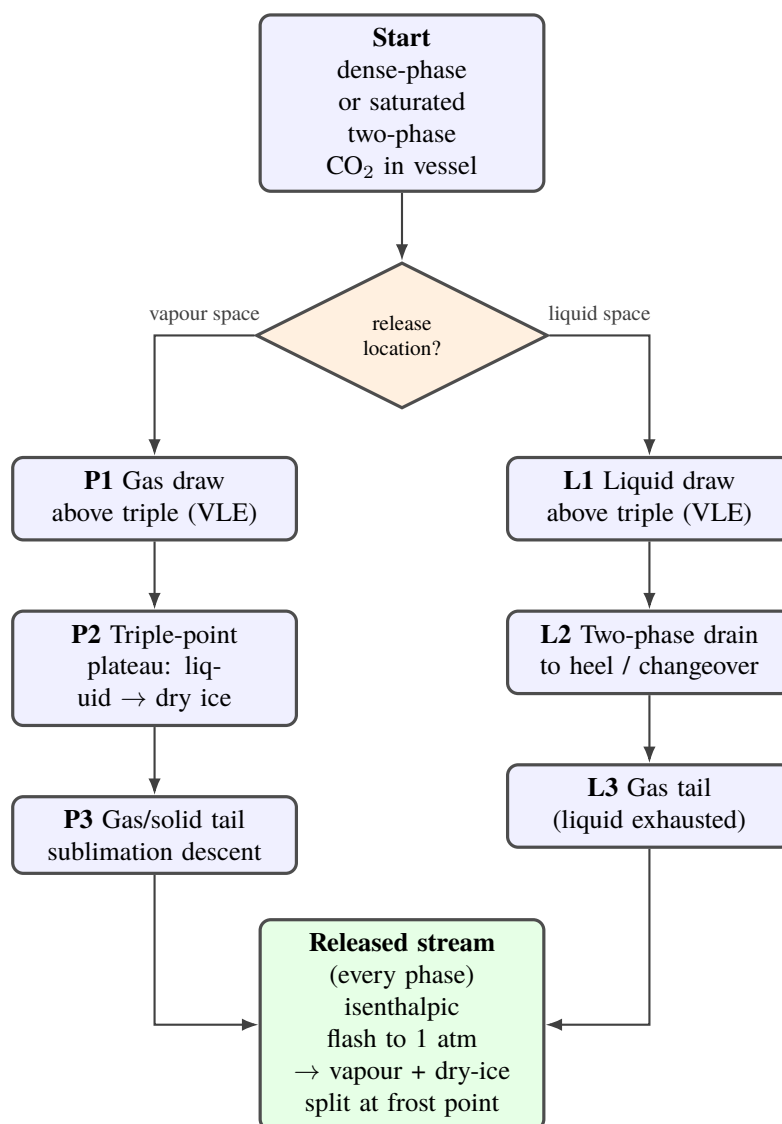


Fig. 2.1: CO₂ release phase decision tree. The left spine (P1–P3) is a vapour-space (gas) release; the right spine (L1–L3) a liquid-space release. The atmospheric dry-ice lever acts on the released stream in every phase.

2.1 Models employed in each phase

Table 2.1: Discharge, vessel (NEM), heat transfer and phase lever by phase

Phase	Discharge	Vessel / NEM	Heat transfer	Phase lever
P1	HEM, saturated-vapour stagnation, $C_{d,gas}$	CoolProp two-zone NEM (gas + liquid), native EoS	two-node wall: gas wall Geankoplis natural convection, wetted wall Rohsenow boiling	vapour/liquid equilibrium (VLE)
P2	HEM gas on the sub-triple throat (forms solid), $C_{d,gas}$	two-zone plateau: warm gas zone + liquid/solid zone at P_{tp}	gas wall Churchill-Chu; Cooper/Rohsenow boiling into the wetted liquid	three-phase invariant lever (m_g, m_l, m_s) from M, V, U
P3	HEM gas on the sublimation line, $C_{d,gas}$	two-zone de-scent: gas leaks/depressurises, dry ice sublimates	gas wall Churchill-Chu; dry-ice wall tracks $T_{sub}(P)$; gas/solid over a mass-derived interface	sublimation-line lever + $\dot{m}_{sub}$ (self-cool + interphase)
L1	HEM liquid + HNE boost $N, C_{d,liq}$	CoolProp two-zone NEM (gas + liquid), native EoS	two-node wall: gas wall Geankoplis natural convection, wetted wall Rohsenow boiling	VLE; liquid boils off as P falls
L2	HEM liquid + $N(P)$ (fades toward the triple point), $C_{d,liq}$	two-zone NEM to the heel; discharge_location sets the gas changeover	two-node wall: gas wall natural convection, wetted wall Rohsenow boiling	VLE; near-adiabatic warm-gas / cold-liquid split
L3	HEM gas, $C_{d,gas}$	residual-gas blow-down (liquid exhausted, negligible in-vessel solid)	gas-contact wall convection	single-phase gas
all	—	—	—	atmospheric lever: isenthalpic flash to 1 atm $\rightarrow$ vapour + dry ice at the frost point

Note

The **triple-point hand-over** ($P1 \rightarrow P2$, or $L2 \rightarrow$ its tail) is where the CoolProp above-triple NEM passes the vessel state to the below-triple two-zone model (*Thermodynamics and the Solid Lever*). The **discharge** model does *not* hand over - the HEM rate is computed on the same CoolProp + solid-table basis at all pressures, so the mass rate is continuous across the triple point.

THERMODYNAMICS AND THE SOLID LEVER

This chapter describes how the add-on evaluates CO₂ properties **entirely from CoolProp** plus a precomputed solid-CO₂ table, the phase regimes, the self-consistent triple point, the lever-rule flashes below it, and the CoolProp hand-overs. The lever/table mathematics is in `co2_release.py` (`CO2ReleaseModel`); the CoolProp-backed implementation is `co2_release_cp.py` (`CO2ReleaseModelCP`), and the solid table is `hyddown.co2_solid`.

3.1 The problem below the triple point

The atmospheric pressure of CO₂ lies below its triple point, so a release crosses into the **vapour + solid** region. CoolProp's reference CO₂ equation of state [4] cannot be used there directly for two reasons:

- it has **no solid phase** - there is no dry-ice branch to flash onto;
- it is **guarded at the triple point** - a normal PropsSI call below 5.18 bar in the two-phase region either raises or snaps to the saturation boundary.

The add-on solves both without a second runtime backend.

3.1.1 CoolProp gas extrapolation (imposed phase)

Pure CO₂ vapour still exists below the triple point (it is the gas in equilibrium with dry ice). CoolProp's low-level `AbstractState` allows the phase to be **imposed**, which makes the Span-Wagner EoS extrapolate smoothly into the metastable gas region past the triple-point guard:

```
self._gas = AbstractState("HEOS", "CO2")
self._gas.specify_phase(CP.iphase_gas)      # forced-gas root below the triple_
↪point
```

Every sub-triple **vapour** property (ρ , u , h , s , and the HEM flux integrand) is taken from this forced-gas state. It was verified against an independent GERG-2008 reference to agree to ~ 0.01 K in temperature and four decimals in phase fraction down to ~ 1 bar, so the extrapolation is quantitatively sound over the whole descent.

3.1.2 The solid-CO₂ table

The one thing CoolProp cannot supply is the **solid** branch. It is provided by a small precomputed table (`hyddown.co2_solid`) of mass-specific solid properties tabulated along the sublimation line $P_{\text{sub}}(T)$:

Table 3.1: Solid-CO₂ table (co2_solid)

Quantity	Role
$P_{\text{sub}}(T), T_{\text{sub}}(P)$ $v_s(T)$	the sublimation line (solid + vapour equilibrium) and its inverse solid specific volume (nearly incompressible; off-line pressure dependence neglected)
$h_s(T), s_s(T)$	solid enthalpy / entropy in CoolProp’s reference basis (so they combine directly with CoolProp fluid values)
$u_s(T, P) = h_s - Pv_s$	solid internal energy
$c_{p,s}(T) = dh_s/dT$	solid heat capacity (for the descent self-cooling term)

The solid values are re-referenced into CoolProp’s enthalpy/entropy basis through the **physical latent heat of sublimation** at each temperature, so a solid property and a CoolProp gas property can be added on one consistent basis - there is no fixed reference-offset mismatch. The table is generated offline (see the provenance note in [Introduction](#)) and committed to the repository; the **runtime uses only CoolProp and numpy**.

3.1.3 Why a table is enough: the single degree of freedom

Below the triple point pure CO₂ is a one-component, two-phase (vapour + solid) system, so by Gibbs’ phase rule it has a **single degree of freedom**: fixing the pressure fixes the temperature ($T_{\text{sub}}(P)$) and both pure-phase states. Every “flash” therefore reduces to **lever-rule arithmetic on the sublimation line** - no solid-aware equation-of-state solver is needed. This lever reproduces a solid-aware reference flash essentially exactly, which is why the table + CoolProp combination is faithful rather than approximate.

3.2 Phase regimes

A CO₂ release passes through three regimes:

1. **Above the triple point** - liquid + vapour equilibrium. The vessel is tracked by the CoolProp non-equilibrium model (NEM); discharge stagnation states are saturated liquid or vapour (native CoolProp).
2. **At the triple point** - solid + liquid + vapour coexist. By Gibbs’ phase rule a pure three-phase point is invariant: T and P are pinned while the leak and heat only shift the phase *amounts* (the freezing “lever”). This is the pressure **plateau** seen in gas-release blowdowns.
3. **Below the triple point** - vapour + solid on the **sublimation line**. Once the liquid is exhausted the state rides $T_{\text{sub}}(P)$ down to the back pressure (frost point ≈ 194.14 K / -79 °C at 1 atm).

3.3 The self-consistent triple point

The literature triple point is $P_{\text{tp}} = 5.18$ bar, $T_{\text{tp}} = 216.59$ K. The solid table is built on the **GERG-2008 self-consistent triple point** used to generate it,

$$T_{\text{tp}} = 216.592 \text{ K}, \quad P_{\text{tp}} = 5.179 \text{ bar}, \quad (3.1)$$

so that the solid and fluid branches meet with no spurious latent-heat jump. The pure-phase specific volumes, enthalpies and internal energies of solid, liquid and vapour at this point ($v_{g,l,s}, u_{g,l,s}$) are stored once as the **vertices of the three-phase lever** (`_init_triple_point`): the solid vertex from the table, the liquid and vapour vertices from CoolProp saturation at P_{tp} . The latent heat of sublimation follows as $L_{\text{sub}} = h_g - h_s$.

3.4 Hand-overs

Two distinct hand-overs occur.

3.4.1 Vessel state: CoolProp NEM → two-zone model (at the triple point)

Above the triple point the vessel inventory is integrated by the CoolProp NEM. Once the tank pressure reaches $P \leq 1.05 P_{\text{tp}}$ and the opt-in `release.solid_in_vessel` is set, the state is handed to the below-triple **two-zone** model (`hdclass.py`). The NEM's two zones - a warm (superheated) gas zone and a liquid/solid zone - carry over their masses and internal energies, and a single-zone ($M_{\text{vessel}}, U_{\text{vessel}}$) state is also seeded for the terminal all-solid regime. From then on the below-triple model (*Dry-Ice Estimation*) advances the vessel.

3.4.2 Discharge: always CoolProp + solid table

The **discharge rate** is computed on the CoolProp + solid-table basis at *all* pressures - above the triple point too - so there is no discontinuity in the mass rate at the triple-point crossing. Only the *vessel state* hands over; the discharge model does not. A fine-resolution check across the triple point shows the rate is smooth and matches a choked-orifice value in both regimes (*Discharge Model*).

Note

The gas *temperature* is not C^1 -continuous across the vessel hand-over: the below-triple descent cools the residual gas a few degrees more steeply than the NEM just above (a $\sim 3^\circ\text{C}$ kink for liquid-release cases). This is small compared with the 0-D stratification gap (*Heat and Mass Transfer*) and is a known, deferred item.

3.5 Precomputed property tables

The below-triple two-zone model steps the gas zone many times with per-step root-finds; live property calls at every node would be slower than necessary. `_init_gas_table` therefore precomputes (once, ~ 1 s) from CoolProp + the solid table:

- a **1-D table** of superheated-gas properties (u, ρ, h , HEM flux G) versus T at the triple-point pressure - used on the plateau, where P is fixed;
- a **2-D table** $(T, P) \rightarrow (\rho, u, h)$ for the descent, where P floats below the triple point (the forced-gas root);
- the **sublimation line** $T_{\text{sub}}(P)$ and the saturated-vapour / solid internal energies along it.

The descent leak rate itself is evaluated live per step (there are far fewer descent steps than table nodes). Interpolants (`gas_T_from_u`, `_gas2d`, `_T_sub_of_P`, ...) provide the state during the integration.

DISCHARGE MODEL

The release rate is a homogeneous-equilibrium (HEM) model with a solid-aware throat, plus an optional non-equilibrium (HNE) boost for a flashing liquid. It is evaluated on the CoolProp + solid-table basis (*Thermodynamics and the Solid Lever*) at all pressures. Implemented in `co2_release.py` / `co2_release_cp.py`.

4.1 Stagnation state

The upstream (stagnation) state feeding the hole follows `release.type` and, once the liquid is exhausted, the phase flip performed in the mass balance (`release_phase liquid → gas`):

- **liquid release** - saturated (or dense) liquid at the tank pressure P_0 (`stagnation(P, "liquid")`);
- **gas release** - saturated vapour at P_0 (`stagnation(P, "gas")`);
- **below the triple point** - an explicit vapour state on the sublimation line (`gas_leak_rate(T, P)`), where a saturated stagnation is undefined.

Each returns the mass-specific stagnation enthalpy h_0 , entropy s_0 , density ρ_0 and temperature T_0 .

4.2 HEM mass flux

The flow is assumed homogeneous (no slip) and isentropic from the stagnation state to the throat. Along the isentrope $s = s_0$, the mass flux at throat pressure P is

$$G(P) = \rho(P) \sqrt{2(h_0 - h(P))}, \quad s(P) = s_0, \quad (4.1)$$

and the discharge maximises G over the throat pressure. The maximiser is the **choked throat**; if G keeps rising down to the back pressure the flow is unchoked and the throat sits at P_{back} :

$$\dot{m} = C_d A G_{\text{max}}, \quad G_{\text{max}} = \max_{P_{\text{back}} \leq P \leq P_0} G(P). \quad (4.2)$$

The isentrope properties come from the CoolProp + solid-table state (`_iso_props`): above the triple point the throat is native CoolProp; below it the throat state is a **vapour + dry-ice** mixture resolved by the sublimation-line lever, and its density/enthalpy carry the solid fraction - the “dry-ice lever at the throat”. Consequently a liquid or gas stream that expands below the triple point correctly forms solid at the orifice.

Note

The flux curve $G(P)$ can carry **two local maxima** across the triple-point kink (one on the vapour+liquid branch above 5.18 bar, one on the vapour+solid branch below it). A bare bounded

search can lock onto the wrong one, so `hem_rate_from_stagnation` first locates the global maximum on a coarse grid and then refines locally (`scipy.optimize.minimize_scalar, bounded`).

The returned dictionary carries `mdot`, `G`, the throat pressure/temperature, the throat solid fraction, a choked flag, and the stagnation state.

Implementation: `co2_release.py:hem_rate()` / `hem_rate_from_stagnation()`.

4.2.1 Verification against theory

At a representative gas point (saturated vapour, 12 bar, 6 mm, $C_d = 0.68$) the CoolProp + solid HEM flux matches a textbook **choked ideal-gas orifice** to within a few percent:

Method	G [kg/m ² s]	$\dot{m}$ [kg/s]
CoolProp + solid HEM (this model)	3878	0.0746
choked ideal-gas orifice	3755	0.0722

So the HEM flux is physically correct; it is the *feed* (which phase, and the effective discharge coefficient) that carries the modelling choices.

4.3 Non-equilibrium liquid discharge (HNE)

A **saturated** liquid discharged through a **short, sharp orifice** has too little residence time to flash to equilibrium: the metastable liquid persists past its saturation pressure and discharges *above* the equilibrium-HEM rate, toward the frozen all-liquid limit [6, 7]. The equilibrium HEM (instantaneous flashing) is the low bound; the frozen incompressible (Bernoulli) flux is the high bound:

$$G_{\text{frozen}} = \sqrt{2 \rho_L (P_0 - P_{\text{back}})}. \quad (4.3)$$

The add-on interpolates between them with a single non-equilibrium factor $N \in [0, 1]$ (`release.liquid_nonequilibrium`):

$$G_{\text{liq}} = \sqrt{(1 - N) G_{\text{HEM}}^2 + N G_{\text{frozen}}^2} \quad (4.4)$$

so $N = 0$ recovers equilibrium HEM and $N = 1$ the frozen liquid. The blend is applied only to a **liquid** stagnation; a gas discharge (single phase, nothing to flash) is untouched. The default is $N = 0$, so existing runs are unchanged. For saturated CO₂ at 10-20 bar the frozen/HEM ratio is large (~ 4), because HEM chokes hard on the low two-phase sound speed while the dense liquid drives a much higher Bernoulli flux; a modest N therefore produces a substantial boost.

This is a deliberately simple metastable interpolation. A fuller non-equilibrium model for CO₂ releases is the Vianna/Lopes HNM [8], derived and validated for **initially sub-cooled / dense-phase liquid at high pressure** (49-159 bar); its dominant incompressible term vanishes at the saturated bubble point, so it is not directly applicable to the CARDICE saturated, low-pressure cases, where the single-factor blend (4.4) is sufficient. The Leung ω -method [9] provides an alternative subcooled/flashing correlation.

Implementation: `co2_release.py:hem_rate()` (liquid branch).

4.4 Orifice and discharge coefficients

The GHGT-16 paper [10] (Table 1) reports the actual restriction-orifice sizes: the pure-CO₂ tests use **3, 4 and 5 mm** holes (Table 4.1; the 5 mm is test 8’s reconciled effective size, see the note below). With the true sizes fixed, the discharge coefficient is split by phase and calibrated directly against the raw 1 Hz Ineris data (*Validation*):

- the **gas** discharge is reliable single-phase HEM (it equals a choked orifice), so it takes one $C_{d,gas}$. Fitting the gas tests (5, 7, 9) gives 0.87-0.92, and the coefficient is *constant along each pressure decline* - an independent confirmation of the HEM + single- C_d gas model. The canonical sharp-edged value $C_{d,gas} = 0.84$ is adopted (low end of the range);
- the **liquid** discharge takes the two-phase / flashing sharp-orifice coefficient $C_{d,liq} = 0.62$ (Darby / API 520 [11]) together with the non-equilibrium factor N .

A liquid release therefore uses $C_{d,liq}$ for the liquid and $C_{d,gas}$ for its gas tail (`release.discharge_coef_gas`).

Table 4.1: Reconciled discharge (true orifices, calibrated to the 1 Hz data)

Test	Phase	P_0 [bar]	Orifice	C_d	N
5	gas	20	3 mm	0.84	–
6	liquid	20	3 mm	0.62	0.174
7	gas	15	4 mm	0.84	–
8	liquid	15	5 mm*	0.62	0.175
9	gas	10	4 mm	0.84	–
10	liquid	10	4 mm	0.62	0.020

* Table 1 lists 4 mm for test 8, but its measured 0.314 kg/s exceeds a controlled 49 bar lab test [12] through the *same* 4 mm hole - impossible, since flow must rise with upstream pressure. An effective 5 mm orifice reconciles it and drops N from 0.53 to 0.18, in line with test 6.

4.4.1 Pressure-dependent non-equilibrium factor

Reconciling N at the fixed $C_{d,liq} = 0.62$ against the measured steady liquid-drain rate shows that, across the saturated-CO₂ regime, N scales linearly with the distance above the triple point:

$$N \approx 0.013 (P_0 - P_{tr}) \quad [\text{bar}], \quad R^2 = 0.88 \quad (4.5)$$

($N \rightarrow 0$ at the triple point, rising to ~ 0.31 at 28 bar). It is applied through `release.liquid_ne_pressure_scaled`, which fades each test’s calibrated N as

$$N_{\text{eff}}(P) = N \cdot \text{clip}\left(\frac{P - P_{tr}}{P_0 - P_{tr}}, 0, 1\right),$$

so $N = N$ at the initial pressure and vanishes near the triple point as the vessel blows down. Physically the metastable boost is a short-residence, above-triple effect: test 10 at 10 bar already reads $\sim$ equilibrium because it sits close to the triple point.

4.5 Hole size, residence time, and full-bore releases

The non-equilibrium boost is a **short-residence** phenomenon: it appears when the liquid does not spend long enough in (and just downstream of) the restriction to nucleate and flash to equilibrium. The main-

stream engineering frameworks parameterise this by **flow length**, not diameter - Fauske's ~ 0.1 m “equilibrium length” [7] and the HNE-DS boiling-delay length threshold both make the flow revert to HEM once the flow path exceeds ~ 0.1 m. The direct CO₂ evidence for a *diameter* effect at fixed length is thin: choked-flow tests through CO₂ orifices and nozzles [13] show HEM's under-prediction shrinking toward larger holes, but confounded by the vena-contracta discharge coefficient, and no clean fixed-length, varying-diameter data exist above ~ 13 mm.

The add-on therefore keeps N **independent of hole diameter** (calibrated per case, faded only with pressure). The responsibility to choose N consistently with the geometry rests with the user, with one important rule:

Warning

Full-bore / large-diameter ruptures: use $C_d = 1$ and $N = 0$ (HEM); do not stack the small-orifice boost on $C_d = 1$. The discharge coefficient (a geometric vena-contracta reduction) and the non-equilibrium factor (a thermodynamic metastability) capture different physics and **anti-correlate with hole size**: a small sharp orifice is low- C_d , high- N (metastable); a full-bore rupture is $C_d \rightarrow 1$, $N \rightarrow 0$ (long flow path, equilibrium). Setting $C_d = 1$ **and** keeping a small-orifice N takes the favourable end of both knobs at once and over-predicts the rate - e.g. at 16 barg saturated liquid, $C_d = 1$ with the calibrated $N \approx 0.175$ gives $\approx 1.9\times$ the $C_d = 1$ HEM rate. Reserve $N > 0$ for genuine small restrictions where $C_{d,\text{liq}} < 1$ is also in play.

4.5.1 Two regimes; HEM vs the HNM

Reconciling N across the wider CO₂ release literature reveals **two regimes that do not share one N law**:

- **low-pressure saturated** CO₂ (10-30 bar; the HydDown regime), where N follows (4.5);
- **high-pressure dense / subcooled** liquid (49-159 bar), where N is governed by subcooling instead.

For the dense regime, plain HEM with a fixed C_d reproduces the measured flux **as well as** the Vianna homogeneous non-equilibrium model (mean absolute error 20.8 % vs 22.2 %), so the HNM/HRM machinery is unnecessary there - HEM plus the low-pressure $N(P)$ boost is sufficient. For **large leaks** the long residence lets the liquid flash to equilibrium, so $N \rightarrow 0$ (pure HEM); the discharge coefficient is geometry-set and essentially size-independent.

DRY-ICE ESTIMATION

Two dry-ice quantities are estimated: the **atmospheric** (downstream) vapour/solid split of the released stream, which is always computed, and the **in-vessel** solid that accumulates when a gas release drags the vessel below the triple point, which is an opt-in model. Atmospheric estimation is in `co2_release.py`; the in-vessel two-zone steppers are dispatched from `hdclass.py`.

5.1 Atmospheric (downstream) dry ice

A free jet does no external work, so the released stream conserves its stagnation enthalpy as it flashes to atmospheric pressure. No solid-aware flash is needed: below the triple point the vapour/solid split is a one-degree-of-freedom lever, so it is computed with an explicit **isenthalpic lever** against pre-computed 1-atm endpoint enthalpies (the pure-vapour $h_{\text{gas}}^{\text{atm}}$ and pure-solid $h_{\text{solid}}^{\text{atm}}$ at the frost point):

$$\beta_{\text{gas}} = \frac{h_0 - h_{\text{solid}}^{\text{atm}}}{h_{\text{gas}}^{\text{atm}} - h_{\text{solid}}^{\text{atm}}}, \quad x_{\text{solid}} = 1 - \text{clip}(\beta_{\text{gas}}, 0, 1) \quad (5.1)$$

If $\beta_{\text{gas}} \geq 1$ the stream is superheated (no dry ice) and a plain gas flash gives its temperature; otherwise the stream is a vapour + dry-ice mixture at the **frost point** $T_{\text{frost}} \approx 194.14 \text{ K}$ (-79°C). The endpoints depend only on the back pressure and are computed once (`_init_atm_endpoints`, from the solid table + `CoolProp`).

An **isentropic** variant (`atm_split_isentropic`) gives the available-work / blast-energy bound. For a liquid release the discharged stream is typically $\sim 50\%$ dry ice by mass at 1 atm - the dry ice a liquid leak produces appears **here**, downstream of the orifice, not inside the vessel.

Implementation: `co2_release.py:atm_split()` / `atm_split_isentropic()`.

5.2 In-vessel dry ice (opt-in)

When a **gas** release keeps the liquid in the vessel, the liquid cools to the triple point, freezes, and a large dry-ice bank is retained; the vessel finally blows down on the sublimation line. This is enabled with `release.solid_in_vessel: true`. Once the tank reaches the triple point (*Thermodynamics and the Solid Lever*) the vessel is advanced by a two-zone model with two regimes: a **triple-point plateau** while liquid remains, and a **sublimation descent** afterwards. A terminal single-zone all-solid step handles the very end when both gas and liquid are gone.

5.2.1 Triple-point plateau (three-phase lever)

At the invariant triple point the temperature and pressure are pinned and the specific volumes and internal energies of the three pure phases ($v_g, v_l, v_s, u_g, u_l, u_s$) are constants. The phase masses then follow from

conservation of total mass, volume and internal energy:

$$\begin{aligned} m_g + m_l + m_s &= M \\ m_g v_g + m_l v_l + m_s v_s &= V \\ m_g u_g + m_l u_l + m_s u_s &= U \end{aligned} \quad (5.2)$$

As the leak removes mass and heat enters, M and U change and the lever redistributes the phases - the liquid steadily converts to solid at fixed T, P . This produces the observed **pressure plateau** at the triple point.

In the **two-zone** plateau (`two_zone_plateau_step`) the gas is carried as a separate warm (superheated) zone that keeps its own temperature - heated by the wall and only weakly coupled to the cold liquid/solid - so the model reproduces the measured gas superheat *and* the freezing lever simultaneously. While liquid remains, the condensate-region wall is **boiling-cooled by the liquid** (Cooper/Rohsenow, *Heat and Mass Transfer*) and that heat enters the liquid/solid zone ($\dot{Q}_{wl}$), boiling some liquid off rather than letting it all freeze. The gas fills the vapour volume at the triple-point pressure.

5.2.2 Sublimation descent

Once the liquid is exhausted the state rides the sublimation line. In the two-zone descent (`two_zone_descent_step`) a warm gas zone leaks and depressurises while the dry-ice bank sublimates only enough to (a) cool itself as $T_{\text{sub}}(P)$ falls and (b) absorb the gas $\rightarrow$ solid interphase heat:

$$\dot{m}_{\text{sub}} = \underbrace{\frac{m_s c_{p,s} (T_s^- - T_s)}{L_{\text{sub}}}}_{\text{self-cooling}} + \underbrace{\frac{\dot{Q}_{gs} \Delta t}{L_{\text{sub}}}}_{\text{interphase}}, \quad \dot{Q}_{gs} = h_{gs} A_{gs} (T_g - T_s). \quad (5.3)$$

Crucially the interphase heat acts over the **gas/solid interface area** A_{gs} - the horizontal cross-section at the top of the settled dry-ice bed, derived from the solid mass and the vessel geometry (`_gas_solid_interface_area`), **not** the much larger gas-wall contact area. Using the bed-top cross-section keeps the well-mixed gas from being dragged down to T_{sub} by an over-large coupling. The interphase HTC h_{gs} (`release.solid_h_gas_solid`, $\approx 3 \text{ W/m}^2\text{K}$) is the single lever that trades gas superheat against retained solid; with it small the dry ice is near-adiabatic and mostly retained. The gas energy balance carries the wall heat, the outflow enthalpy and the sublimed-vapour enthalpy; the solid mass is decremented by $\dot{m}_{\text{sub}}$. The dry-ice-region wall tracks $T_{\text{sub}}(P)$ (*Heat and Mass Transfer*).

5.2.3 Two-zone parameters

Parameter	Meaning
<code>solid_h_gas_wall</code>	below-triple gas $\leftrightarrow$ wall HTC; "churchill" (default, Churchill-Chu), "calc" (Geankoplis), or a fixed number (<i>Heat and Mass Transfer</i>).
<code>solid_h_inner</code>	plateau wall $\leftrightarrow$ boiling-liquid HTC; "cooper" (default), "calc" (Rohsenow), or a fixed number.
<code>solid_h_gas_solid</code>	descent gas $\leftrightarrow$ dry-ice interphase HTC over the bed-top interface area (trades gas superheat vs retained solid; $\approx 3 \text{ W/m}^2\text{K}$).
<code>solid_h_gas_liquid</code>	plateau gas $\leftrightarrow$ liquid interphase HTC (small - keeps the gas warm; ≈ 0).
<code>solid_gas_wall_frac</code>	optional fixed override of the gas-contact wall fraction (default: from geometry).
<code>discharge_location</code>	liquid level (m) at which a liquid draw switches to the gas tail (heel changeover).

5.3 Behaviour

The model reproduces the central experimental contrast:

- **Gas releases** (tests 5/7/9) retain a large in-vessel dry-ice bank ($\sim 266/302/392$ kg modelled vs $\sim 217/298/412$ kg measured) with a multi-hour triple-point plateau.
- **Liquid releases** (tests 6/8/10) retain essentially none: the liquid drains and boils out **before** the vessel reaches the triple point, so it never freezes in place. The dry ice appears downstream instead ((5.1)).

See *Validation* for the full comparison.

HEAT AND MASS TRANSFER

The two-phase (non-equilibrium) vessel carries **separate gas and liquid zones** with their own temperatures, and a **two-node wall** (a gas-contact node and a condensate “wetted” node). This chapter documents the wall model above and below the triple point, the convection coefficients, the initialisation, and the zone mass/energy balances. Implemented in `hdclass.py` (with correlations in `transport.py` and the below-triple wall in the two-zone steppers).

6.1 Two-node wall above the triple point (NEM)

Above the triple point the CoolProp non-equilibrium model resolves the wall into an **unwetted (gas-contact)** node T_{wall} and a **wetted (liquid-contact)** node T_{wall}^w , exchanging with the respective fluid zone:

$$\begin{aligned}\dot{Q}_{\text{inner}} &= h_{\text{in}} A_{\text{unwetted}} (T_{\text{wall}} - T_g) \\ \dot{Q}_{\text{inner}}^w &= h_{\text{in}}^w A_{\text{wetted}} (T_{\text{wall}}^w - T_l)\end{aligned}\tag{6.1}$$

with the wetted/unwetted areas from the vessel geometry at the current liquid level. The inner coefficients follow natural-convection correlations (`h_inner: calc`); the gas side uses the **gas** temperature in the NEM. The external side uses `h_outer` (the Jamois-measured $0.39 \approx 0.4 \text{ W/m}^2\text{K}$, *Experimental Basis*).

6.2 Natural convection (Geankoplis)

The above-triple natural-convection HTC uses the Geankoplis [14] Nusselt correlation with the Rayleigh number $\text{Ra} = \text{Gr Pr}$:

$$\text{Gr} = \frac{g \beta |T_{\text{wall}} - T_{\text{fluid}}| L^3}{\nu^2}, \quad \text{Nu} = \begin{cases} 0.13 \text{Ra}^{1/3} & \text{Ra} \geq 10^9 \\ 0.59 \text{Ra}^{1/4} & 10^4 < \text{Ra} < 10^9 \\ 1.36 \text{Ra}^{1/5} & \text{otherwise} \end{cases} \quad h = \frac{\text{Nu } k}{L} \tag{6.2}$$

with L the vessel height (vertical) or diameter (horizontal) and fluid properties (β, ν, k, Pr) from CoolProp.

6.3 Below the triple point (two-zone wall)

Below the triple point the two-zone model carries the wall as two nodes, and the physics switches between the **plateau** (liquid present) and the **descent** (dry ice only).

6.3.1 Gas-contact wall (both regimes)

The gas-contact wall exchanges with the gas and the ambient (the condensate is adiabatic to it):

$$m_w c_{p,w} \frac{dT_{\text{wall}}}{dt} = \dot{Q}_{\text{out}} - \dot{Q}_{wg}, \quad \dot{Q}_{wg} = h_{gw} A_g (T_{\text{wall}} - T_g), \quad \dot{Q}_{\text{out}} = h_{\text{out}} A_{\text{out}} (T_{\text{amb}} - T_{\text{wall}}), \tag{6.3}$$

where A_g is the gas-contact area from the vessel geometry and the condensate volume, and the wall thermal mass / outer area are partitioned between the two nodes by area fraction so the total is not double-counted.

6.3.2 Plateau wetted wall: boiling into the liquid

While **liquid** is present the condensate-region wall is **boiling-cooled** by the liquid, not gas-heated. The wall $\rightarrow$ liquid heat uses a nucleate pool-boiling coefficient (below) and flows **into** the liquid/solid zone, boiling some liquid off:

$$m_w^w c_{p,w} \frac{dT_{\text{wall}}^w}{dt} = \dot{Q}_{\text{out}}^w - \dot{Q}_{wl}, \quad \dot{Q}_{wl} = h_{\text{boil}} A_{\text{wet}} (T_{\text{wall}}^w - T_{\text{tp}}). \quad (6.4)$$

The wall relaxes toward the cold boiling liquid pinned at the triple point, and $\dot{Q}_{wl}$ enters the plateau lever (*Dry-Ice Estimation*) as extra vaporisation - the warm wall's stored sensible heat boils liquid off rather than letting it all freeze.

6.3.3 Descent dry-ice wall: sublimation-tracking

Once the liquid is exhausted the condensate-region wall is buried against the cold dry-ice bed and **tracks the sublimation temperature**:

$$T_{\text{wall}}^w = T_{\text{sub}}(P) \quad (\text{while dry ice is present}). \quad (6.5)$$

No wall $\rightarrow$ solid heat is applied (the ice bed holds the wall on the sublimation line), so the retained dry-ice mass is unaffected by the wall. This ice-pinned wall reproduces the CARDICE bottom-sensor cool-down to $\approx -75^\circ\text{C}$.

6.3.4 Phase-depletion clamp

When a phase is depleted, its wall node has no fluid to exchange with, so it **clamps to the surviving phase's wall** (the condensate wall clamps to the gas wall once the dry ice is gone). This avoids a stranded node drifting to an unphysical temperature.

6.4 Churchill-Chu gas-wall convection

The default below-triple gas $\leftrightarrow$ wall coefficient (`release.solid_h_gas_wall: "churchill"`) is the Churchill-Chu [15] free-convection correlation for a vertical plate - the same law the Høydalsvik/Munkejord reference model [16] uses for the vapour region:

$$\text{Nu} = \left(0.825 + \frac{0.387 \text{Ra}_{L_v}^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right)^2, \quad h_{gw} = \frac{\text{Nu } k}{L_v}. \quad (6.6)$$

Two features matter. First, the properties are the CO₂ **vapour** properties at the film temperature, taken from CoolProp's imposed-gas root (valid below the triple point). Second, the characteristic length is the **vapour-column height** $L_v = (1 - \alpha_\ell) L_z$ above the settled condensate, which shrinks as dry ice and liquid accumulate; for a sphere/horizontal cylinder (`length = 0`) the vessel diameter is used. The single-equation Churchill-Chu form is smooth across the laminar-turbulent transition, unlike the piecewise Geankoplis form (6.2), which remains available as "calc". In the turbulent regime the two agree; the choice only matters in the low-Rayleigh tail. A fixed number is also accepted.

No special-casing of a small gas/wall ΔT is needed. As $\Delta T \rightarrow 0$ the Churchill-Chu correlation tends to its **conduction floor** ($\text{Nu} \rightarrow 0.825^2$, i.e. $h_{gw} \rightarrow 0.68 k/L_v$) - a finite, physically-correct small value, not a singularity - and the Geankoplis form tends to zero; in either case the heat rate $h_{gw} A_g \Delta T$ vanishes there regardless. The coefficient therefore returns the correlation value at all ΔT , and falls back to a nominal 15 W/m²K only on a non-finite CoolProp film state.

6.5 Nucleate boiling on the wetted wall

The plateau wall $\rightarrow$ liquid coefficient (`release.solid_h_inner`) is a nucleate pool-boiling correlation evaluated at the triple-point saturated liquid:

- "cooper" (default) - the Cooper [17] reduced-pressure correlation, which carries CO₂'s high reduced pressure natively (the law the SINTEF reference model uses);
- "calc" - the Rohsenow [18] correlation with a CO₂-on-steel surface constant ($C_{sf} = 0.013$, $n = 1.7$);
- a fixed number.

Cooper is the default for a physical reason. Its reduced-property form,

$$h_{\text{boil}} = 55 P_r^{0.12-0.2 \log_{10} R_p} (-\log_{10} P_r)^{-0.55} M^{-0.5} q^{0.67}, \quad (6.7)$$

needs only the **reduced pressure** $P_r = P/P_c$, the molar mass M , the wall heat flux q and the surface roughness R_p . It requires **no surface tension, no saturated liquid/vapour densities and no latent heat** - exactly the properties that are ill-defined for CO₂ at and below the triple point, where no stable liquid exists and the metastable saturated-liquid state is poorly characterised. The Rohsenow form needs all of them ($\sigma, \rho_\ell, \rho_g, h_{fg}, c_{p,\ell}, \mu_\ell$), which must be evaluated at the pinned triple-point saturated liquid and carry that extrapolation uncertainty. Cooper sidesteps it with quantities that are well defined at the triple-point reference.

Both are capped at 3000 W/m²K: above that the wall is conduction-limited and its temperature is insensitive to the exact coefficient (a sweep on the CARDICE cases confirms the retained mass and wall temperature barely move across `cooper/calc/fixed`). This conduction-limited insensitivity is also why the exact boiling law is not critical - it sets how *fast* the wall cools, not the sensible-heat budget that fixes the retained dry ice. The descent uses no boiling law - the dry-ice wall is sublimation-pinned (above).

6.6 Gas-liquid interphase

`calculation.h_gas_liquid` accepts "calc"/"calc_two_sided" (natural-convection correlations across the interface) or a fixed number. For a liquid drain a small value keeps the warm gas decoupled from the cold boiling liquid (near-adiabatic), reproducing the measured warm-gas / cold-liquid split. The descent gas $\leftrightarrow$ dry-ice interphase is separate and acts over the **bed-top interface area** (*Dry-Ice Estimation*), not the wall area.

6.7 Initialisation of the warm gas and split wall

Two related initialisation features (both keyed on `initial.gas_temperature`):

- **Superheated gas zone.** With `initial.gas_temperature` the gas zone starts warmer than the saturated liquid (a lower density and mass; the fill takes up the difference at a matched total mass). The value is set to the **mid-band (median) of the top 3 gas-space thermocouples** at t_0 - a representative warm-gas start rather than the single warmest sensor.
- **Split wall init.** With a superheated gas zone the wall has equilibrated with the phase it touches before the blowdown, so the **gas-contact wall starts at the gas temperature** and the **wetted wall at the liquid temperature** (rather than both at `initial.temperature`). An optional `initial.wall_temperature` gives a uniform override. Default behaviour (no `gas_temperature`) is unchanged.

6.8 Zone mass and energy balances (NEM)

Each zone integrates its own internal energy. The outflow enthalpy is **attributed to the phase the mass actually leaves from**: a liquid-space release draws liquid, a vapour-space release / standard blowdown draws gas (exactly one of h_g^{out} , h_l^{out} is non-zero):

$$\begin{aligned}\Delta U_g &= -\dot{m} h_g^{\text{out}} \Delta t + \dot{Q}_{\text{inner}} \Delta t - \dot{Q}_{gl} \Delta t + E_{\text{evap}} - E_{\text{cond}} \\ \Delta U_l &= -\dot{m} h_l^{\text{out}} \Delta t + \dot{Q}_{\text{inner}}^w \Delta t + \dot{Q}_{gl} \Delta t - E_{\text{evap}} + E_{\text{cond}}\end{aligned}\quad (6.8)$$

where $\dot{Q}_{gl} = h_{gl} A_{\text{interface}} (T_g - T_l)$ is the interphase heat and E_{evap} , E_{cond} are the boil-off / condensation phase-transfer energies handled in the mass balance. Both wall $\rightarrow$ gas and wall $\rightarrow$ liquid terms are present, so the wall heats both zones.

6.9 The gas-temperature gap (stratification)

For the **gas** releases the modelled 0-D gas node runs colder than the measured warm top (e.g. ~ -49 vs -27°C for test 5). The real gas is **stratified** - warm on top, cold near the interface - which a single 0-D gas node cannot represent; it averages the two and mixes in cold boil-off vapour. This is a fundamental 0-D limitation, not a heat-transfer-coefficient error (the wall model reproduces the measured wall, and the gas-wall HTC is the physical free-convection value). Capturing it would need a stratified (multi-node) gas zone; Vessfire, another 0-D-gas tool, shows the same gap [19].

Fig. 6.1 shows the mechanism for a gas release: the gas expansion-cools during the fast blowdown, then **re-warms** during the constant-pressure plateau as the wall heats the near-isobaric gas, while the two wall nodes separate into a warm gas-contact wall and a cold dry-ice-contact wall.

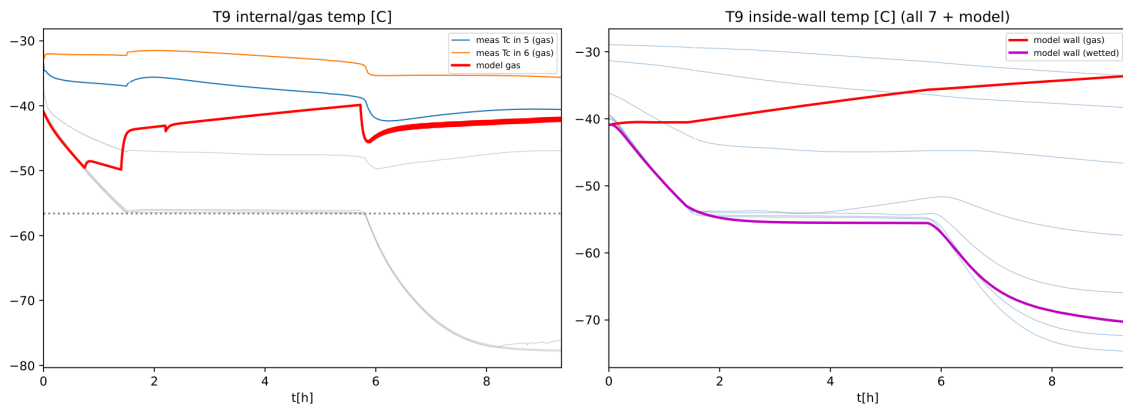


Fig. 6.1: Test 9 gas-release diagnostic: internal/gas temperatures (left) and inner-wall sensors vs the two modelled wall nodes (right).

EXPERIMENTAL BASIS

The add-on is validated against **two independent** CO₂ blowdown campaigns that together span the pressure range of interest:

- **CARDICE** (Ineris, 2 m³ sphere) - low/medium pressure (10-20 bar), saturated two-phase, both gas- and liquid-space releases; the source of the in-vessel dry-ice physics.
- **Høydalsvik/Munkejord** (SINTEF/NTNU, 0.06 m³ vertical cylinder) - high pressure (120 bar), dense-phase, both no-riser (gas-space) and riser (liquid-draw) releases; a finely instrumented cross-check of the discharge, dry-ice and stratification behaviour.

7.1 CARDICE (Ineris 2 m³ sphere)

The **CARDICE** (CARbon Dioxide ICE) Joint Industry Project, in which Ineris performed pilot-scale CO₂ blowdown experiments in a 2 m³ sphere. Two papers document the work: the GHGT-15 paper [19], which gives the test matrix and the Vessfire comparison, and the IJGGC setup paper [20], which documents the rig, instrumentation and calibration in detail. The open 1 Hz dataset is on Zenodo [21].

7.2 The vessel

The vessel characteristics reported by [20] are summarised in Table 7.1.

Table 7.1: Vessel characteristics

Property	Value
Geometry	sphere, two welded hemispheres
Internal volume	2.030 ± 0.005 m ³ (geometric)
Inner radius / diameter	778 mm / 1556 mm
Mean wall thickness	54 ± 5 mm
Total vessel mass	4.80 t (steel shell ≈ 4 t + 3 flanges)
Steel grade	P355GH (1.0473)
Heat capacity c_p	450 J/(kg·K)
Thermal conductivity λ	44 W/(m·K)
Density	7800 kg/m ³
Design	200 bara / −60 °C

In HydDown the sphere is modelled as two hemispherical heads with zero cylinder length (vessel.type: Hemispherical, length: 0, diameter: 1.58). The wall thickness is set to 0.072 m rather than the physical 54 mm so that the shell-corrected wall mass matches the measured 4.80 t total (the extra accounts for the three heavy flanges/nozzles, which HydDown does not model separately).

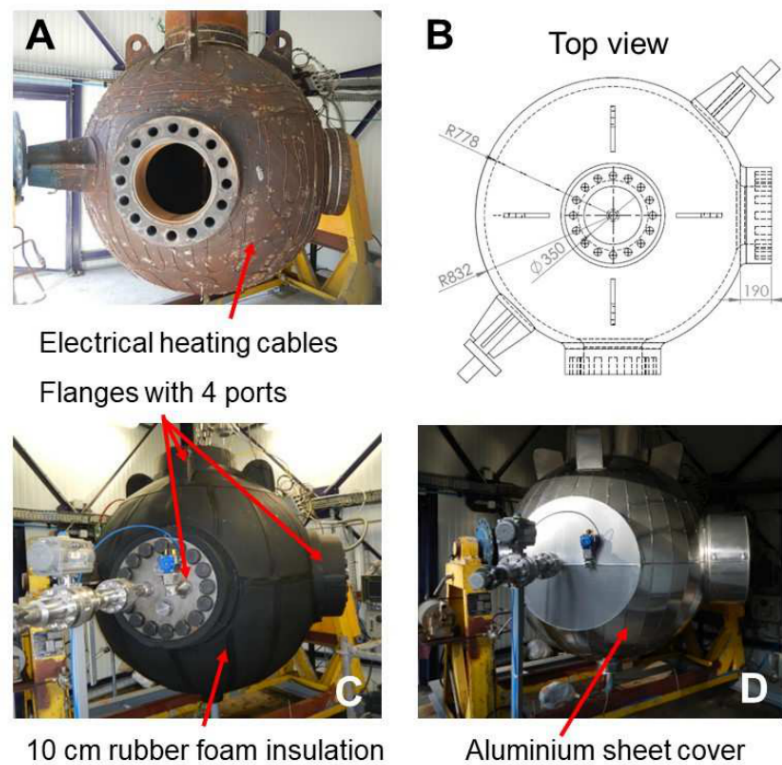


Figure 1: The 2m³ steel vessel. (A) bare, (B) Top view drawing, (C) with the and the insulation, (D) with aluminium sheet cover

Fig. 7.1: The 2 m³ steel sphere: (A) bare, (B) top-view drawing, (C) with flanges and insulation, (D) with the aluminium cover.

From [20].

7.3 Insulation and external heat transfer

The sphere is insulated with 10 cm of rubber foam ($\lambda_{fo} = 0.034 \text{ W/(m}\cdot\text{K)}$) over an outer area of 11 m^2 . [20] reports the external heat transfer in two ways:

Theoretical (steady conduction through the foam):

$$h_{\text{theo}} = \frac{\lambda_{fo} A_{\text{out}}}{L_{fo}} = \frac{0.034 \times 11}{0.10} = 3.75 \text{ W/K} \quad (7.1)$$

Measured (a cooling calorimetry test, their Fig. 2): the insulated vessel was warmed and allowed to cool towards the 2°C ambient. The contents dropped $\Delta T = 8^\circ\text{C}$ over $\Delta t = 45 \text{ h}$. The heat-loss rate is inferred from the vessel thermal mass C (dominated by the 4 t of steel, $4000 \times 450 = 1.8 \text{ MJ/K}$, giving $\approx 89 \text{ W}$; $\approx 107 \text{ W}$ including the contents):

$$\dot{Q} = C \frac{\Delta T}{\Delta t} \approx 107 \text{ W}, \quad h = \frac{\dot{Q}}{\Delta T} = \frac{107}{25} = 4.3 \text{ W/K} \quad (7.2)$$

where $\overline{\Delta T} \approx 25^\circ\text{C}$ is the mean inside-outside gap over the run. Note that this h is a **whole-vessel conductance in W/K**, not a per-area coefficient; per unit outer area it is $4.3/11 = 0.39 \text{ W/(m}^2\cdot\text{K)}$, which is the value used as `heat_transfer.h_outer` in the models. The measured 4.3 W/K exceeds the pure-foam 3.75 W/K by $\approx 15\%$ because the real heat paths also include the flanges, the release pipe, and the thermocouple wells ($8 \text{ mm} \times 250 \text{ mm}$ pipes protruding through the insulation) - thermal bridges the conduction estimate ignores.

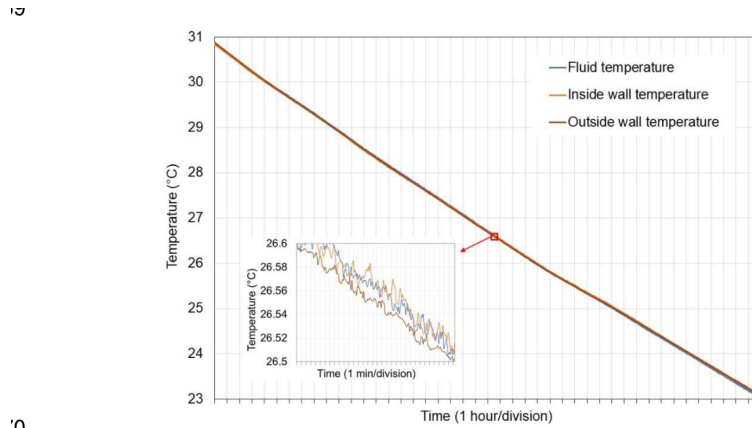


Figure 2: Temperature curves recorded during the cooling of the insulated vessel from 31°C to ambient (outside temperature = 2°C)

Fig. 7.2: Cooling curves of the insulated vessel used to obtain the measured external heat-transfer conductance.

From [20].

For the several-hour CO_2 blowdowns this $\sim 100 \text{ W}$ ingress is small against the latent (flashing/sublimation) loads, so the blowdown is effectively near-adiabatic.

7.4 Instrumentation

The measured quantities are:

- absolute **pressure** inside the sphere and in the release pipe;
- **fluid temperature** at 6 heights inside the sphere ($T_{c1} \dots T_{c6}$) and in the pipe;
- inner and outer **wall temperature** at 7 heights ($T_{c1} \dots T_{c7} / T_{cF1} \dots T_{cF7}$);

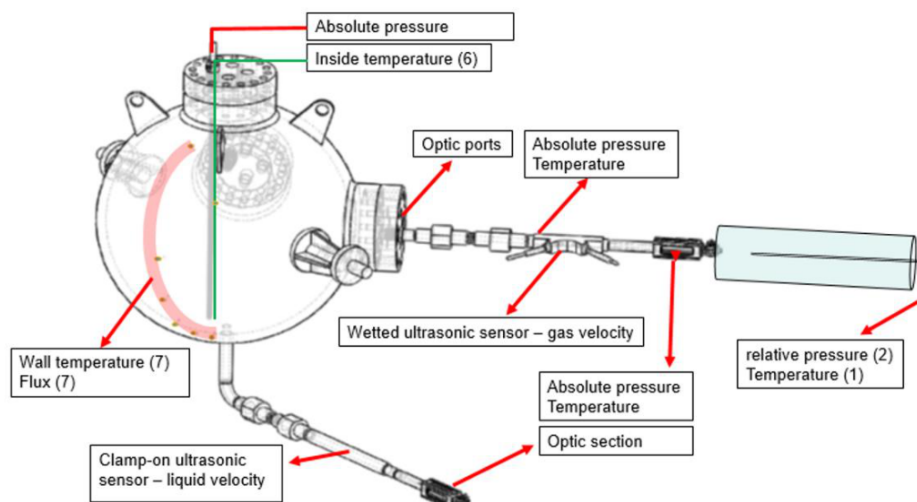


Fig. 2. Overall instrumentation of the experimental set-up.

Fig. 7.3: Overall instrumentation of the set-up.
From [19].

- **mass flow rate** through the release pipe;
- the **mass of the vessel** (fluid + solids) via 4 load cells (Mettler-Toledo 0745A, 2200 kg each, ± 100 g; total-mass uncertainty ± 0.4 t on 4 t; flow-rate uncertainty ± 5 %).

Two independent flow measurements were used: direct **weighing** (load cells) and an **ultrasonic** velocity (inline on the gas pipe, clamp-on on the liquid pipe) combined with the computed density. The load-cell mass, corroborated against the measured temperature and pressure, is the reference used here for the initial inventory and the discharge-rate calibration. A camera and a katharometer were also fitted.

The release pipe is 40 mm inner diameter, ~ 2 m long, terminated by a **calibrated orifice** connected to the vessel middle (vapour release) or bottom (liquid release). The paper gives one pure-CO₂ orifice example of **4 mm** (range 1 mm to full bore); per-test sizes are not reported, so they are inferred here from the measured discharge (*Discharge Model*).

7.5 Test matrix

The pure-CO₂ tests are 5-10; tests 1-4 are methane / methane-CO₂ mixtures and are out of scope. For the low-temperature tests the sphere was filled with liquid to about the middle. The matrix is reported by [19].

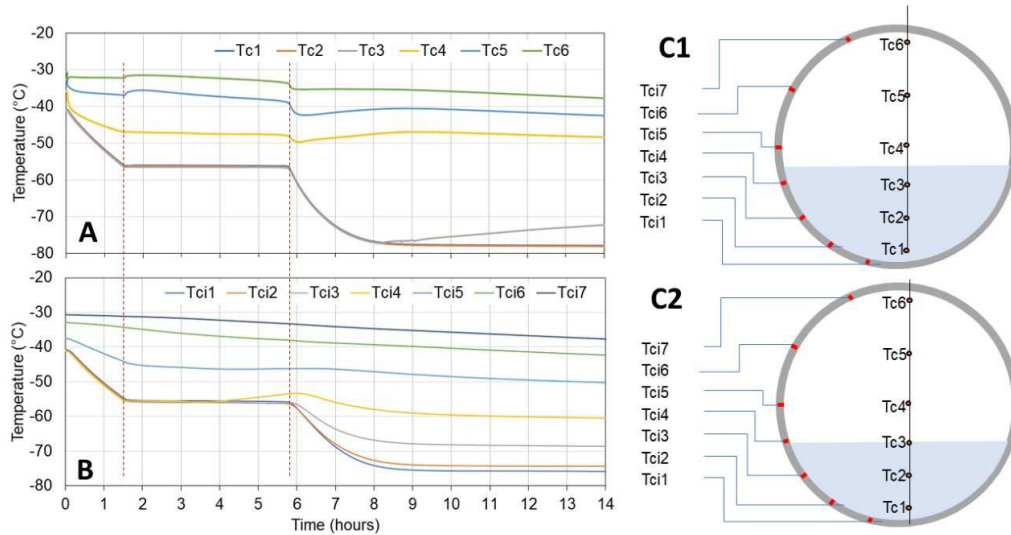


Figure 16: Temperature profile of the fluid. (A) inside the vessel (Tcx) and (B) in the inner wall surface (Tcix) during the release. On the right side, drawings of the vessel showing (C1) the location of the gas/liquid interface at $t=0$ (C1) and (C2) the

Fig. 7.4: Example measured temperatures during a blowdown: (A) fluid at 6 heights inside the vessel and (B) the wall 5 mm from the inner face, showing the stable vertical stratification of the vapour phase. From [20].

Table 7.2: CARDICE pure- CO_2 test matrix

Test	Composition	P_0 [bar]	T_0 [°C]	Release	HydDown file
5	100 % CO_2	20	-20	Gas	CARDICE_test5.yml
6	100 % CO_2	20	-20	Liquid	CARDICE_batch6.yml
7	100 % CO_2	15	-29	Gas	CARDICE_test7.yml
8	100 % CO_2	15	-29	Liquid	CARDICE_test8.yml
9	100 % CO_2	10	-40	Gas	CARDICE_test9.yml
10	100 % CO_2	10	-40	Liquid	CARDICE_test10.yml

Gas releases produced large in-vessel dry-ice banks (over 450 kg reported), as in Fig. 7.5; liquid releases produced essentially none inside the vessel.

7.6 Raw-data alignment

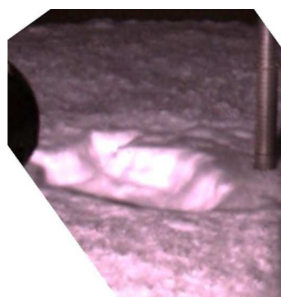
The 1 Hz files include a pre-release period; the true blowdown start t_0 is detected from the onset of the inventory-mass decline (and the pressure drop) and all comparisons are time-shifted to it (e.g. $t_0 = 438$ s for cardice-06). Folder cardice-0N corresponds to test N .



Fig. 3. Solid CO₂ formed inside the sphere during a gas release test. The picture is tilted, the vertical is indicated by the metal rod in contact with dry ice.

Fig. 7.5: Solid CO₂ formed inside the sphere during a **gas** release test (the picture is tilted; the metal rod indicates the vertical).

From [19].



596
597 Figure 19: Close-up showing the dry ice foam-like structure. Image extracted from a
598 video taken several hours after complete vaporization of the liquid phase
599

Fig. 7.6: Close-up of the dry-ice “foam-like” structure (floating solid with vapour bubbles).

From [20].

7.7 Høydalsvik/Munkejord (SINTEF dense-phase cylinder)

The second campaign is the SINTEF/NTNU dense-phase depressurization experiments [16], which release CO₂ from a small, heavily instrumented vertical cylinder at ~ 120 bar. Relative to CARDICE it probes a very different part of the envelope - **dense-phase (sub-cooled) start, high pressure, thick relative wall** - and its dense fluid flashes hard through the nozzle, so it stresses the discharge and flash-boiling models rather than the multi-hour in-vessel dry-ice bank. The open dataset is on Zenodo (record 19589510, CC BY 4.0).

7.7.1 The vessel

The vessel is a SS316 cylinder with an inner diameter of 273.0 mm and an internal height of 1000 mm, a 25.4 mm wall and a 50 mm bottom plate, closed by an 80 mm lid (580 mm diameter) on an 83 mm flange and standing on three legs. Its design pressure is 138 bar. It is insulated with 50 mm Armaflex LTD (walls/bottom) and a 45 mm inflatable lid cover.

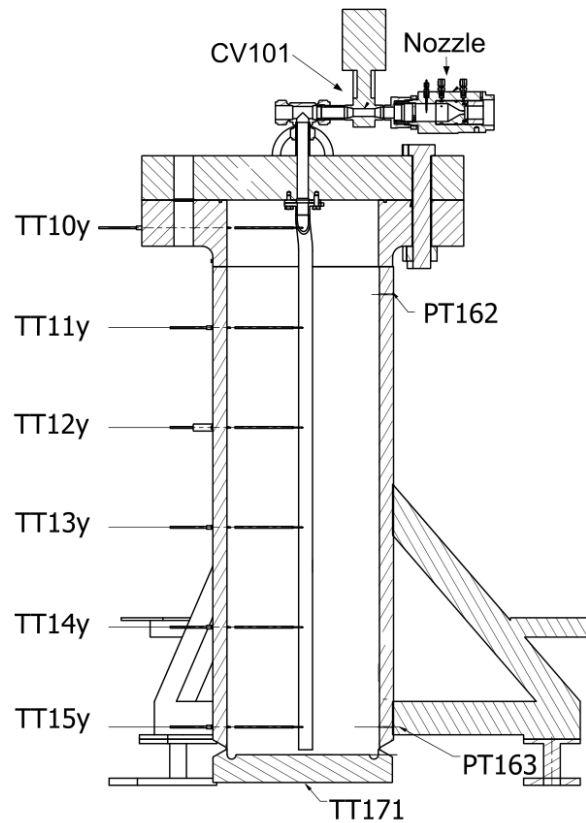


Fig. 1. Schematic of the vessel experimental setup in the ECCSEL depressurization facility.

Fig. 7.7: Schematic of the vessel experimental setup in the ECCSEL depressurization facility: the cylindrical vessel with the removable riser tube, the six fluid/wall thermocouple heights (TT10y-TT15y), the vessel pressure sensors PT162 (top) / PT163 (bottom), the bottom outer-wall sensor TT171, and the nozzle with valve CV101 on the lid.

From [16].



Fig. 7.8: Picture of the insulated vessel in the ECCSEL depressurization facility.
From [16].

Table 7.3: SINTEF vessel characteristics (from Høydalsvik et al.)

Property	Value
Geometry	vertical circular cylinder, three legs
Inner diameter / internal height	273.0 mm / 1000 mm
Internal volume	$\approx 0.058 \text{ m}^3$
Wall / bottom / lid thickness	25.4 / 50 / 80 mm
Steel grade (ρ , k , c_p)	SS316 (7950 kg/m ³ , 16.3 W/m/K, 500 J/kg/K)
Design pressure	138 bar
Insulation	50 mm Armaflex LTD + 45 mm lid cover
Wall-to-bore ratio t/D	≈ 0.093 (vs ≈ 0.046 for the CARDICE sphere)

In HydDown the cylinder is modelled as `vessel.type: Flat-end`, `orientation: vertical`, `length: 1.0`, `diameter: 0.273`, `thickness: 0.0254`. The thicker *relative* wall matters below the triple point: it feeds more lateral heat into the dry-ice-contact wall than the CARDICE geometry (*Validation*).

7.7.2 Release configurations

CO₂ is discharged through a converging nozzle in one of two configurations:

- **no-riser (gas-space) release** - the nozzle draws from the top of the vessel; the dense liquid stays in the vessel, giving the gas-draw / in-vessel dry-ice behaviour (the P1-P3 spine of *Model Map (Phase Decision Tree)*);
- **riser (liquid-draw) release** - a 20 mm riser tube draws liquid from ≈ 9 mm above the floor to the lid centre (set in the model by `release.discharge_location`), so the vessel drains as a liquid release and empties before reaching the triple point.

The CO₂ leaves through valve CV101 and a converging nozzle of throat diameter **8.0, 6.5 or 4.5 mm** (inlet diameter 32 mm, half-angle $\theta = 60^\circ$), matching the choked-flow orifice/nozzle study [13] on the same facility.

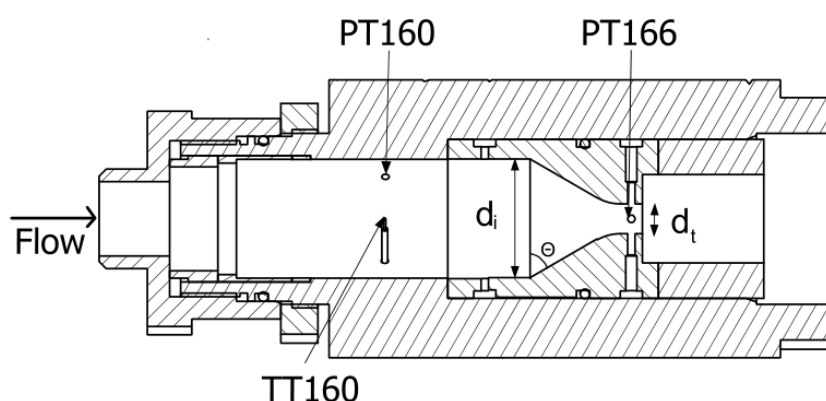


Fig. 7.9: Schematic of the interchangeable converging nozzle, with the inlet diameter d_i , throat diameter d_t and half-angle θ , and the throat pressure sensors PT160 / PT166.

From [16].

7.7.3 Instrumentation

The vessel is densely instrumented with **26 thermocouples, four Keller PA33Xe pressure sensors and three scales** (one per leg, for the vessel weight). Fluid and wall temperatures are measured at **6 heights** (TT10y at 950 mm down to TT15y at 50 mm from the bottom plate) and, per height, at four radial positions - labelled TT1xy with x the height and y the radial position: y = 4 fluid on the central axis, y = 3 fluid 5.5 mm from the wall, y = 2 wall 3 mm inside the inner face, y = 1 outer wall. Vessel pressure is the mean of PT162 (170 mm from the top) and PT163 (50 mm from the bottom); PT160 / PT166 sit just before and at the nozzle throat. Sampling is at 1000 Hz, averaged to 0.5 Hz.

In the validation the fluid and wall temperatures are shown as **upper / lower bands** (the top three vs the bottom three sensor heights) to bracket the vertical stratification, and each channel is read on its own time base to avoid a clock mismatch between the pressure, temperature and weight streams. The blowdown onset is taken from the dense-flash pressure drop and all traces are shifted to it.

7.7.4 Test matrix

The nine pure-CO₂ cases used here are:

Table 7.4: Høydalsvik/Munkejord test matrix (as modelled)

Test	P_0 [bar]	T_0 [°C]	Nozzle	Configuration
Exp71	122.6	25.2	8.0 mm	no-riser (gas)
Exp72	119.0	24.9	6.5 mm	no-riser (gas)
Exp75	119.0	25.0	4.5 mm	no-riser (gas)
Exp52	119.9	15.4	8.0 mm	riser (liquid)
Exp53	119.5	24.4	8.0 mm	riser (liquid)
Exp56	119.1	15.2	6.5 mm	riser (liquid)
Exp57	116.8	24.5	6.5 mm	riser (liquid)
Exp45	119.9	14.5	4.5 mm	riser (liquid)
Exp46	116.7	24.4	4.5 mm	riser (liquid)

VALIDATION

The CO₂ add-on is validated against the pure-CO₂ **CARDICE** set (Ineris, tests 5-10) and the **Høy-dalsvik/Munkejord** dense-phase set (SINTEF, 9 cases), both described in *Experimental Basis*. Each case has a tracked input file and a four-panel comparison figure; the generators are scripts/validate_cardice_reconciled.py and scripts/validate_munkejord.py.

8.1 CARDICE (Ineris)

8.1.1 Methodology

For every test:

1. **Time base** - the measured trace is aligned to the model at a mid-blowdown **reference pressure** (P fallen $\sim 15\%$ of its span), which is robust for both the fast gas cases and the slow liquid drains; acquisition gaps in the 1 Hz files (e.g. a 385 s outage in test 10) are merged and the temperature step across the gap removed.
2. **Initial mass** - vessel.liquid_level is calibrated (via the CoolProp initial phase masses) to the measured load-cell inventory; all six match within 0.4 %.
3. **Discharge** - true 3/4/6 mm orifices [10], a phase-split C_d ($C_{d,gas} = 0.84$, $C_{d,liq} = 0.62$) and the liquid HNE boost N (*Discharge Model*).
4. **Below the triple point** - the gas wall uses solid_h_gas_wall: churchill (Churchill-Chu), the plateau wetted wall solid_h_inner: cooper (Cooper boiling), and the descent dry-ice wall tracks the sublimation line (*Heat and Mass Transfer*).
5. **Heat transfer** - h_outer = 0.4 W/m²K (Jamois).

8.1.2 Master comparison

Table 8.1: Model vs measured (CARDICE tests 5-10, current model)

Test	Phase	m_0 mdl/meas [kg]	Retained dry ice mdl/meas [kg]	Wetted (bottom) wall min mdl/meas [°C]
5	gas	824 / 823	266 / 217	-78 / -75
6	liquid	804 / 802	0 / 4.2	-24 / -24
7	gas	790 / 791	302 / 298	-76 / -75
8	liquid	761 / 758	0 / 3.8	-32 / -32
9	gas	904 / 904	392 / 412	-72 / -75
10	liquid	745 / 743	0 / 5.1	-42 / -40

The central physics is reproduced across the set: **gas releases retain 266-392 kg of in-vessel dry ice with multi-hour triple-point plateaus and walls to -75°C ; liquid releases retain essentially none, with warm walls** - the dry ice a liquid leak makes appears downstream instead. Test 7 lands on the measured retained mass; test 5 over-predicts (its 4.8-t wall stores the most sensible heat) and test 9 slightly under-predicts. The liquid tests empty to a residual gas heel rather than the measured $\sim 4\text{-}5\text{ kg}$, a small non-retained end-effect.

8.1.3 Per-test comparisons

Each figure has four panels: vessel pressure; inventory mass (with the modelled gas / liquid / solid breakdown); the internal fluid thermocouples as **two bands** (upper three = gas, lower three = liquid/solid) vs the modelled gas and liquid/solid temperatures; and the inner-wall thermocouples as two bands (upper = gas wall, lower = wetted wall) vs the modelled gas-contact and wetted wall nodes.

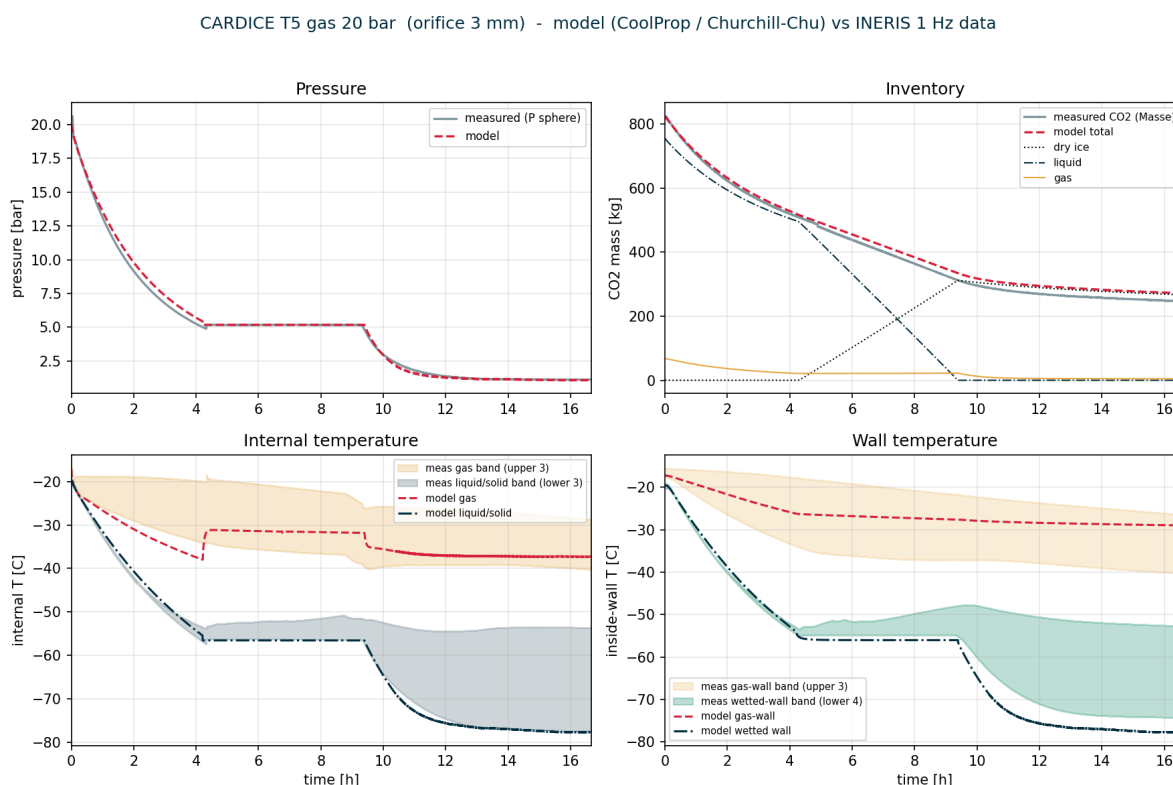


Fig. 8.1: Test 5 - gas release, 20 bar. Long triple-point plateau, $\sim 266\text{ kg}$ retained dry ice, wetted wall to -78°C .

8.1.4 Benchmark against Vessfire (Test 5)

Test 5 is the paper's benchmark. [19] report the measured pressure, mass flow rate, phase temperatures and wall temperatures against Vessfire (Fig. 8.7). HydDown reproduces the same three-stage behaviour - gas/liquid blowdown, triple-point plateau, gas/solid sublimation descent - and, like Vessfire, captures the pressure and rate well while under-predicting the (stratified) gas temperature.

CARDICE T6 liquid 20 bar (orifice 3 mm) - model (CoolProp / Churchill-Chu) vs INERIS 1 Hz data

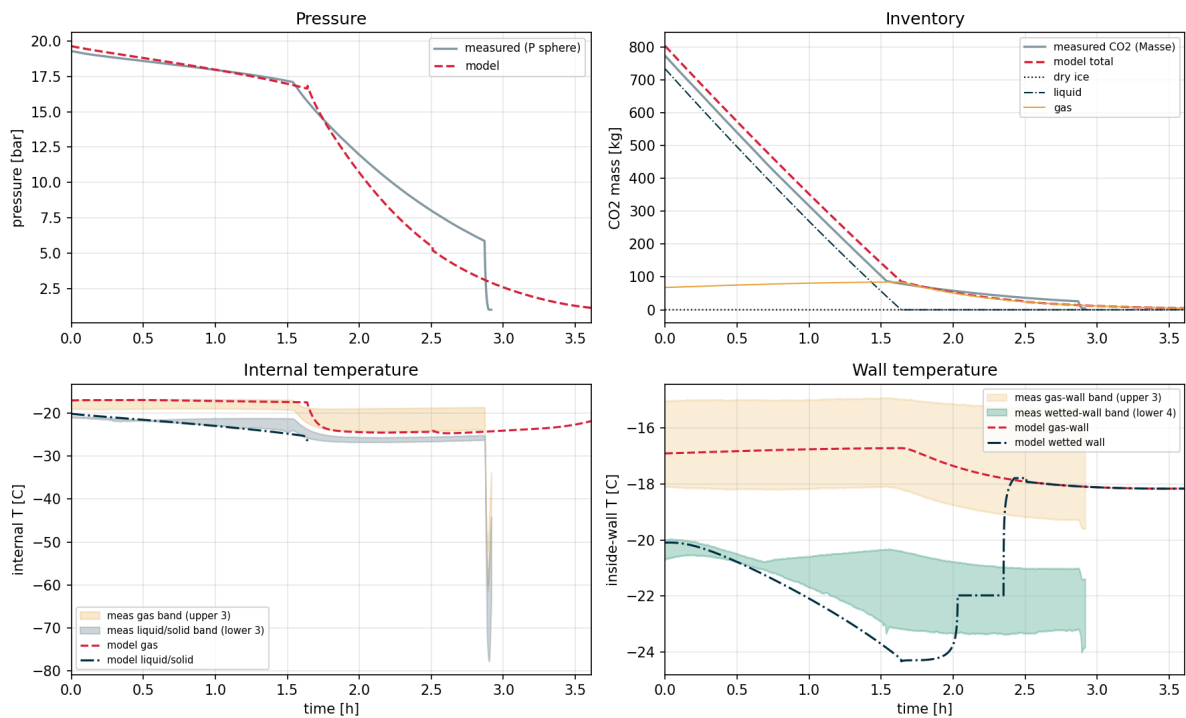


Fig. 8.2: Test 6 - liquid release, 20 bar. Liquid drains and boils out before the triple point; no in-vessel dry ice.

CARDICE T7 gas 15 bar (orifice 4 mm) - model (CoolProp / Churchill-Chu) vs INERIS 1 Hz data

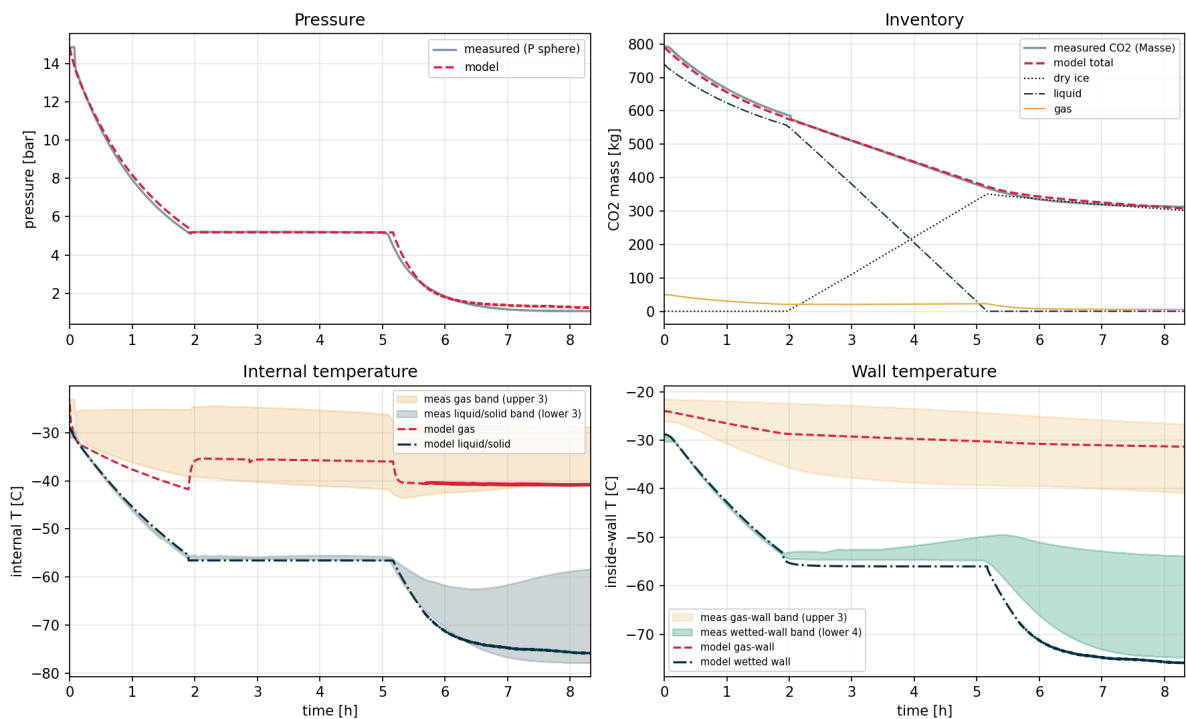


Fig. 8.3: Test 7 - gas release, 15 bar. Retained dry ice on the measured value (302 vs 298 kg).

CARDICE T8 liquid 15 bar (orifice 5 mm) - model (CoolProp / Churchill-Chu) vs INERIS 1 Hz data

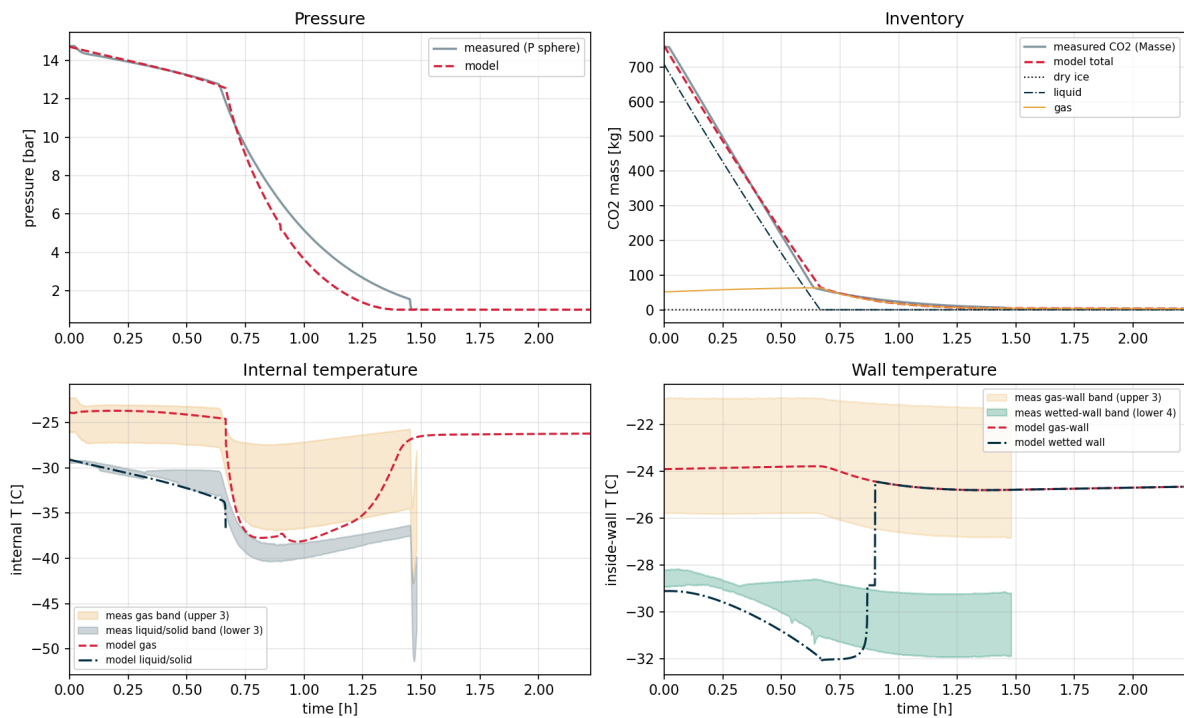


Fig. 8.4: Test 8 - liquid release, 15 bar.

CARDICE T9 gas 10 bar (orifice 4 mm) - model (CoolProp / Churchill-Chu) vs INERIS 1 Hz data

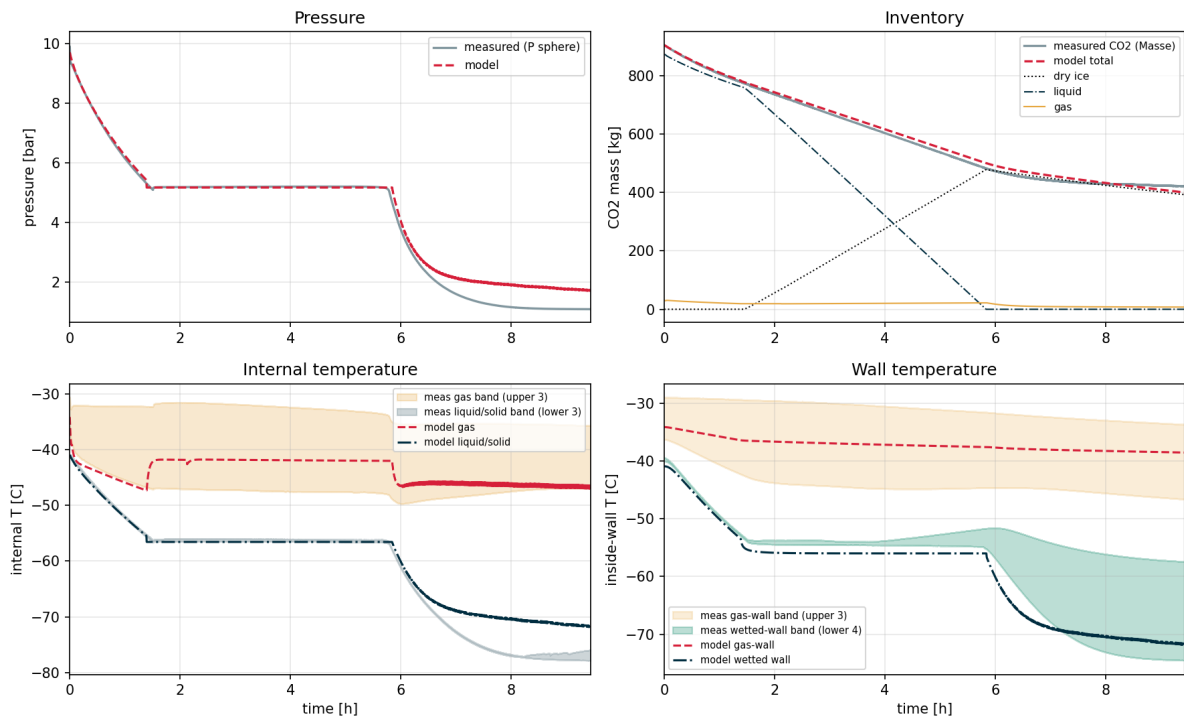


Fig. 8.5: Test 9 - gas release, 10 bar. The most retained dry ice (~ 392 kg).

CARDICE T10 liquid 10 bar (orifice 4 mm) - model (CoolProp / Churchill-Chu) vs INERIS 1 Hz data

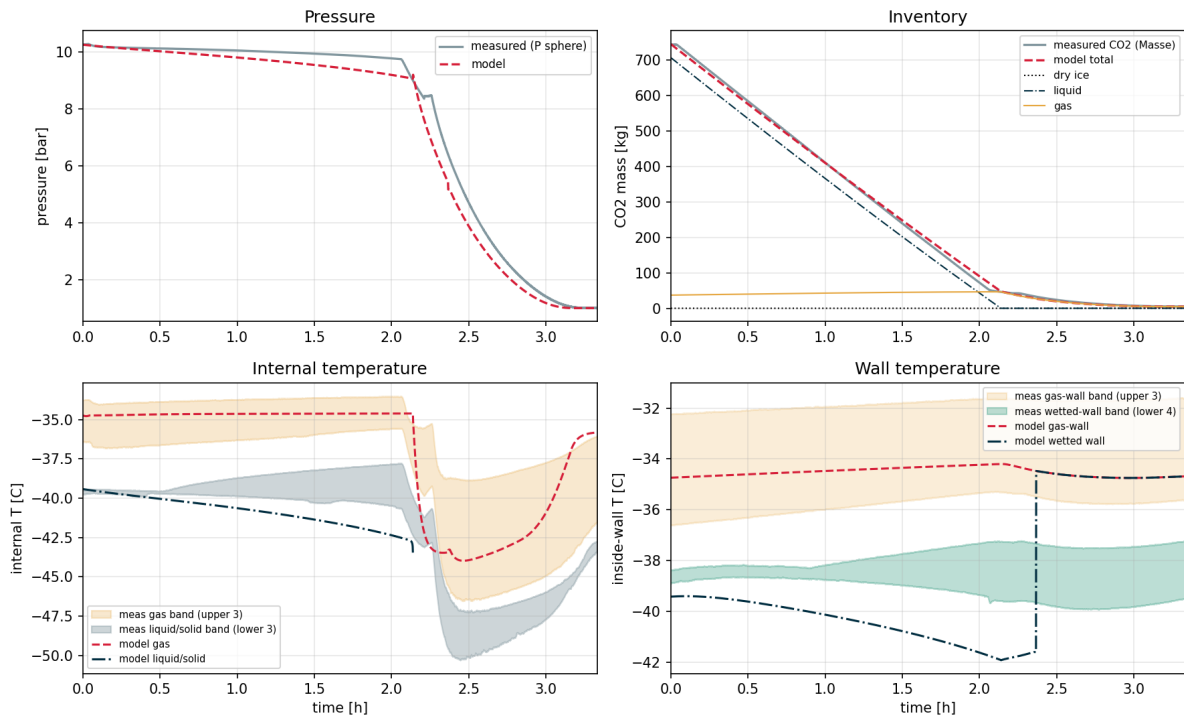


Fig. 8.6: Test 10 - liquid release, 10 bar. The 385 s acquisition gap at ~ 2.2 h is merged in the measured trace.

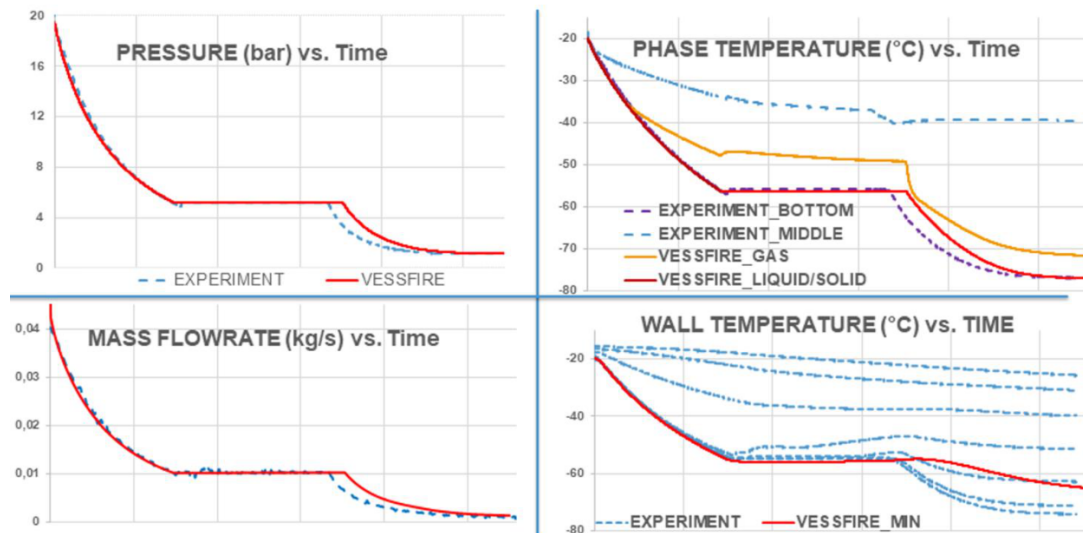


Fig. 5. Test 5 – Experimental vs. simulation results.

Fig. 8.7: Test 5 experimental vs Vessfire results (pressure, mass flow rate, phase temperature, wall temperature).

From [19].

8.2 Høydalsvik/Munkejord (SINTEF)

The dense-phase cases are modelled with the same physics (CoolProp + solid table, HEM + N , Churchill-Chu / Cooper walls) with the vertical-cylinder geometry and, for the riser cases, `discharge_location` at the 9 mm riser inlet (*Experimental Basis*). Fluid and wall temperatures are compared as **upper / lower bands** against the model gas and liquid/solid (and gas-wall / wetted-wall) nodes.

The no-riser (gas) cases retain a small in-vessel dry-ice bank that the model tracks closely: $\sim 8.4 / 7.8 / 8.1$ kg modelled for Exp71 / 72 / 75, against a measured ~ 8.4 kg for Exp71 (a clean match on the flagship case). The riser (liquid) cases drain and empty to essentially zero in-vessel solid, as measured. Across all nine the model's two zones bracket the measured stratification envelope.

Munkejord Exp71 (123 bar, 25.2 C, 8.0 mm, no-riser / gas draw) - model (CoolProp) vs 1 Hz data

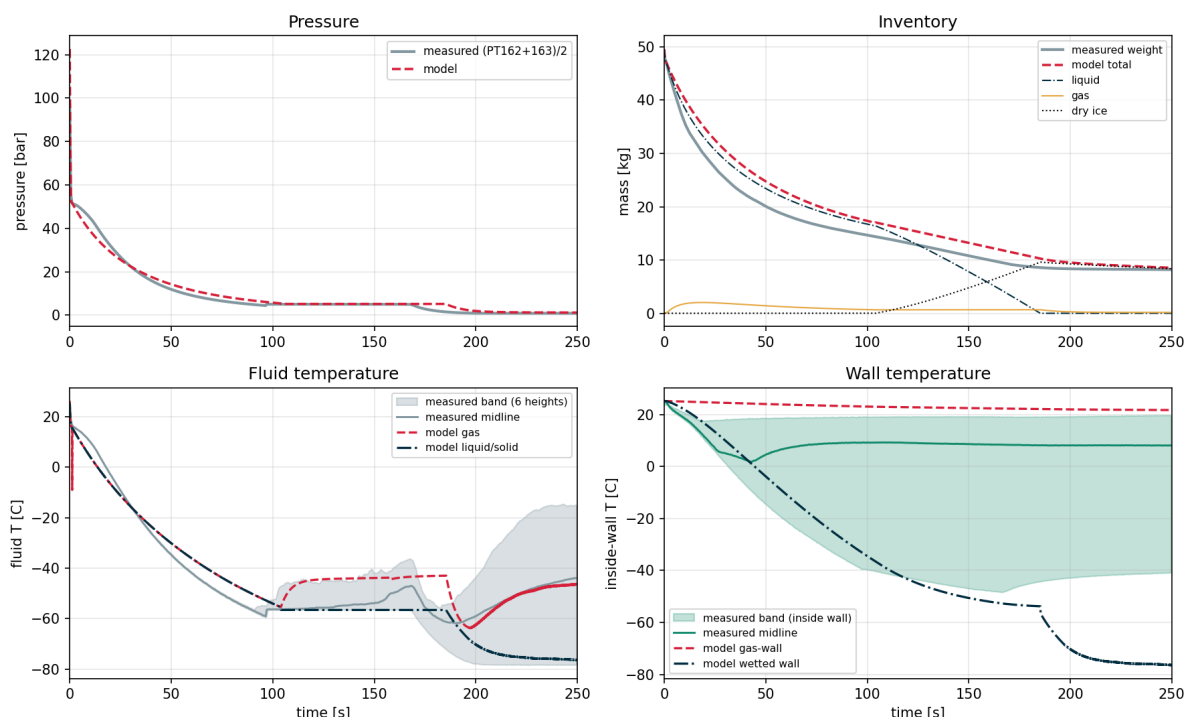


Fig. 8.8: Exp71 - no-riser (gas) release, 122.6 bar, 8.0 mm. Pressure, inventory (with dry-ice breakdown), fluid-temperature band and wall-temperature band vs the model.

(The remaining riser cases Exp45, Exp52 and Exp56 behave analogously and are generated by `scripts/validate_munkejord.py`.)

8.3 Remaining gaps

These are documented model limitations, not defects:

- **Gas-temperature stratification.** The 0-D gas node runs colder than the measured warm top on the gas cases; a single node cannot hold the vertical gradient (*Heat and Mass Transfer*). Vessfire, a 0-D-gas tool, shows the same [19].
- **Dry-ice wall: CARDICE vs Munkejord.** In the gas/solid tail the dry-ice-region wall is sublimation-pinned, which matches CARDICE's thick ice bed (walls to -75°C) but reads too cold for the Munkejord geometry, where the measured bottom wall re-warms (thin ice, relatively

Munkejord Exp72 (119 bar, 24.9 C, 6.5 mm, no-riser / gas draw) - model (CoolProp) vs 1 Hz data

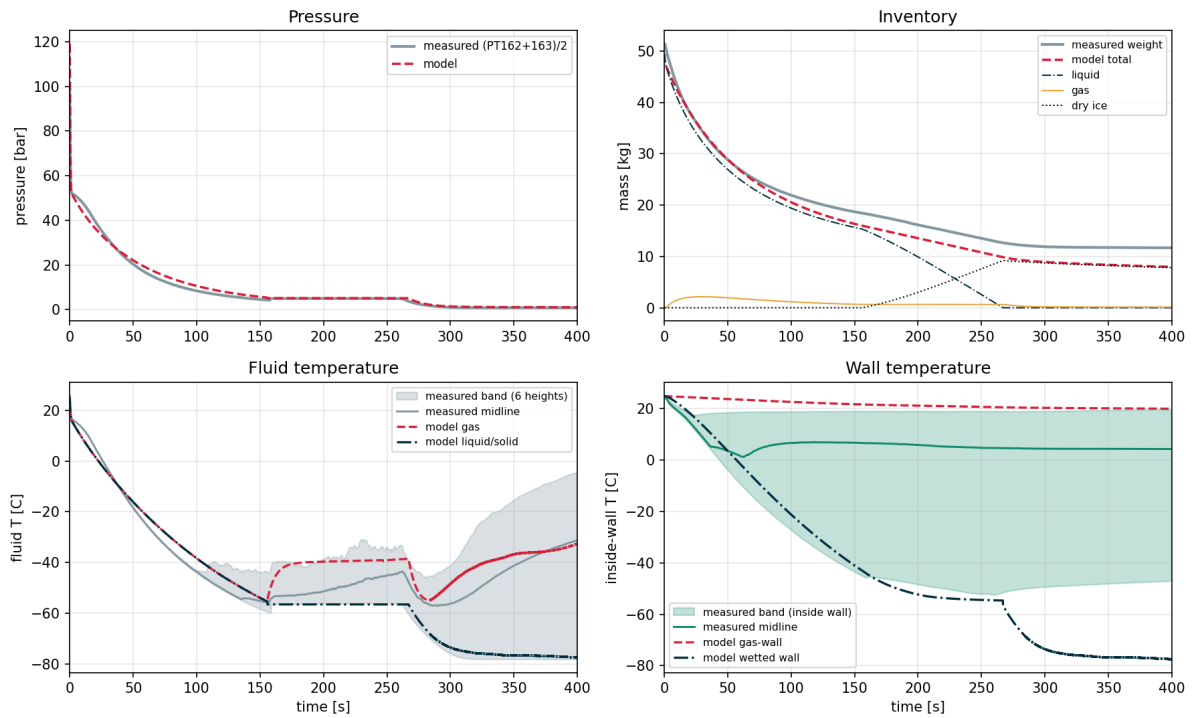


Fig. 8.9: Exp72 - no-riser (gas) release, 119 bar, 6.5 mm.

Munkejord Exp75 (119 bar, 25.0 C, 4.5 mm, no-riser / gas draw) - model (CoolProp) vs 1 Hz data

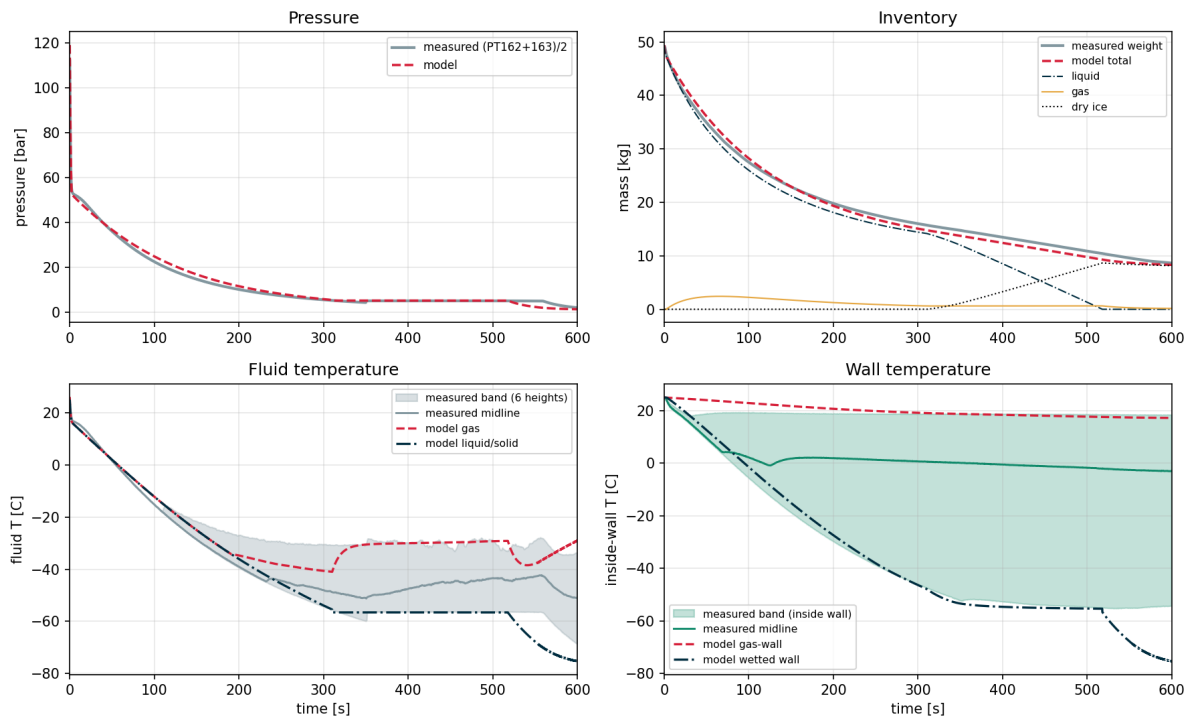


Fig. 8.10: Exp75 - no-riser (gas) release, 119 bar, 4.5 mm.

Munkejord Exp53 (120 bar, 24.4 C, 8.0 mm, riser / liquid draw) - model (CoolProp) vs 1 Hz data

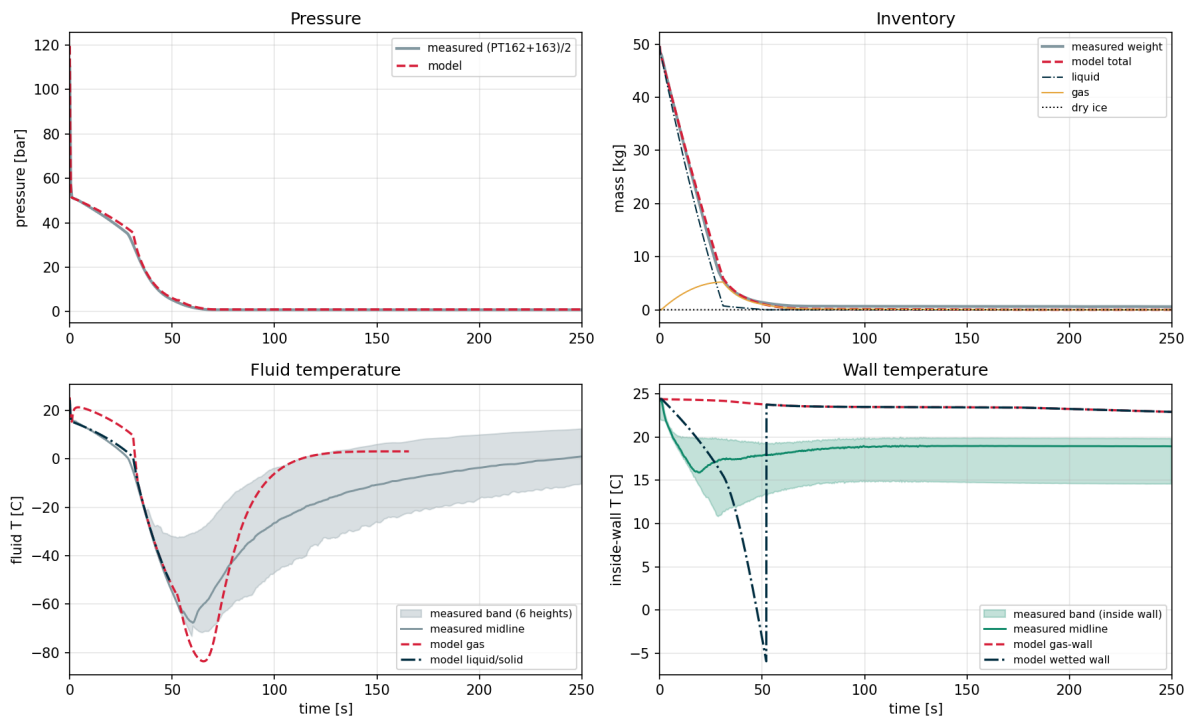


Fig. 8.11: Exp53 - riser (liquid) release, 119.5 bar, 8.0 mm. The liquid draws through the riser and the vessel empties before the triple point.

Munkejord Exp57 (117 bar, 24.5 C, 6.5 mm, riser / liquid draw) - model (CoolProp) vs 1 Hz data

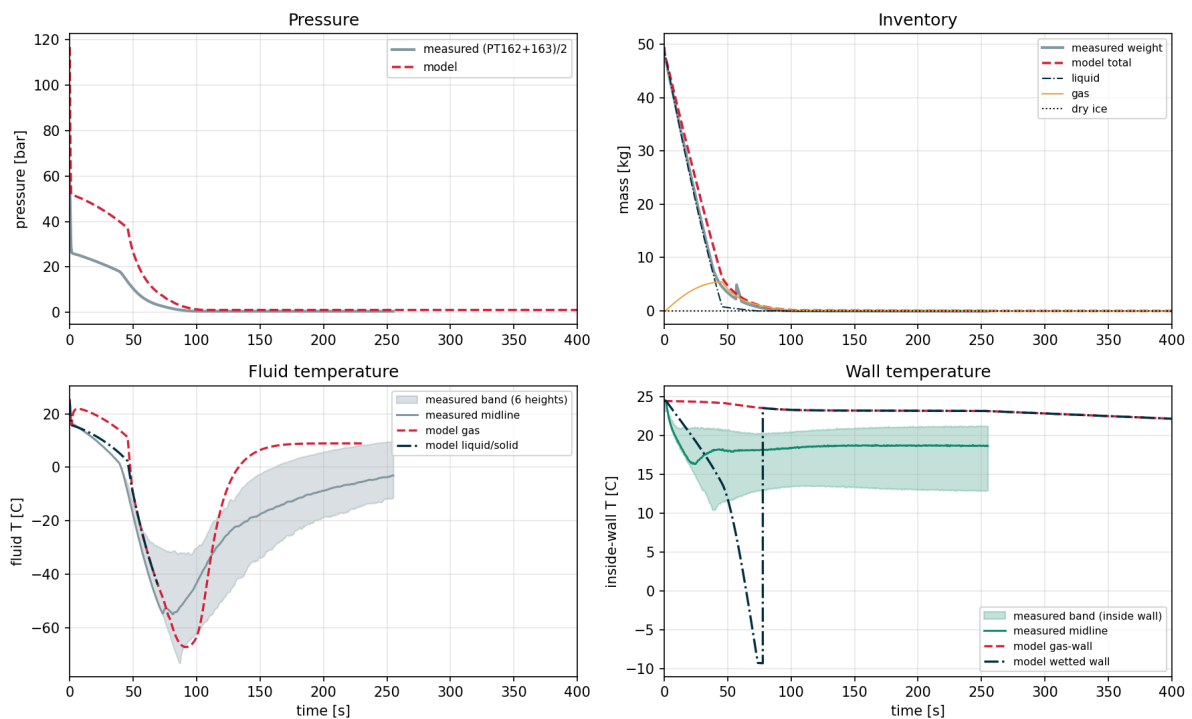


Fig. 8.12: Exp57 - riser (liquid) release, 116.8 bar, 6.5 mm.

Munkejord Exp46 (117 bar, 24.4 C, 4.5 mm, riser / liquid draw) - model (CoolProp) vs 1 Hz data

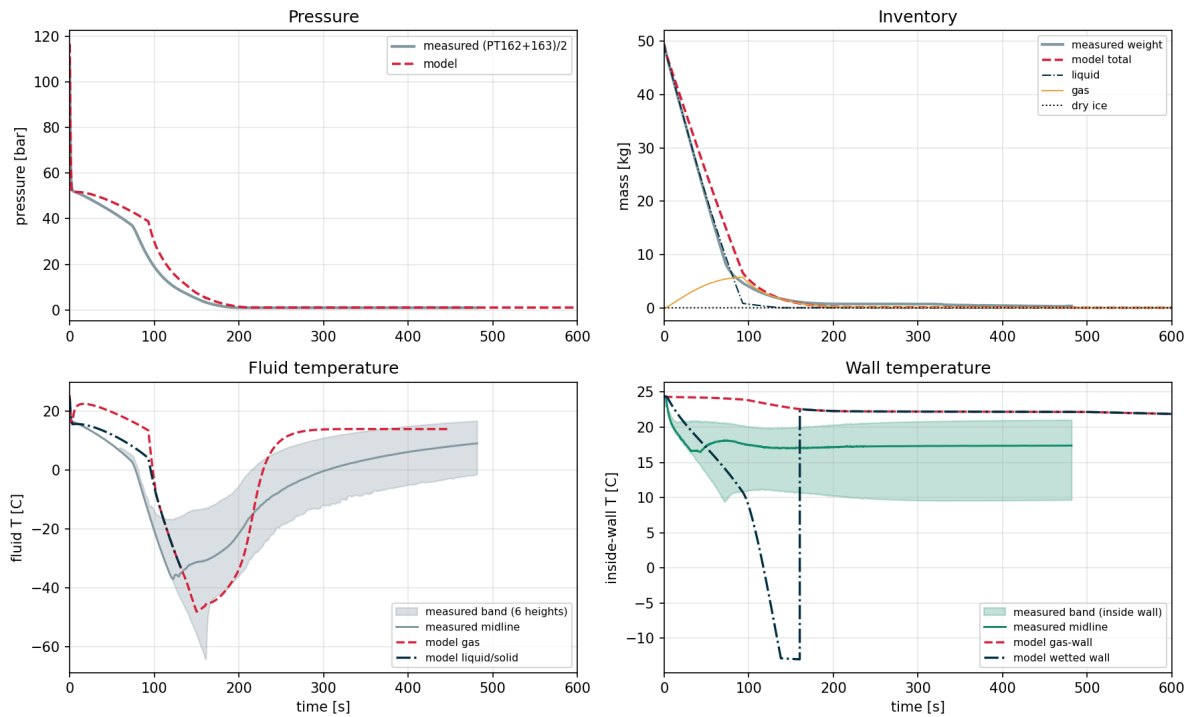


Fig. 8.13: Exp46 - riser (liquid) release, 116.7 bar, 4.5 mm.

thicker wall, so lateral 2-D wall conduction feeds heat back into the ice-contact metal). A phenomenological lateral-conduction lever to span both is a deferred item.

- **Triple-point handover kink.** A $\sim 3^\circ\text{C}$ gas-temperature kink at the CoolProp above-triple $\rightarrow$ below-triple vessel hand-over (liquid cases); small vs the stratification gap; deferred.
- **Liquid-release heel.** The liquid cases empty to a residual gas heel rather than the measured $\sim 4\text{-}5$ kg (CARDICE) - a small non-orifice end-effect a steady discharge model does not reproduce.

8.4 Regression

The base HydDown behaviour is unchanged: all new inputs default off (`liquid_nonequilibrium: 0`, no `gas_temperature`), and `test_all.py` passes except pre-existing, unrelated failures (a Tk/plotting environment issue and a modified `rupture.yml` example). A dedicated `src/hyddown/test_co2_release.py` covers the CO₂ module.

NOMENCLATURE AND INPUT SCHEMA

9.1 Principal symbols

Symbol	Meaning	Units
G	mass flux (per unit throat area)	kg/m ² /s
$\dot{m}$	mass flow rate	kg/s
C_d	discharge coefficient ($C_{d,gas}$, $C_{d,liq}$)	–
N	non-equilibrium (HNE) factor, 0 = HEM, 1 = frozen	–
$P_0, T_0, \rho_0, h_0, s_0$	stagnation pressure / temperature / density / enthalpy / entropy	Pa, K, kg/m ³ , J/kg, J/kg/K
P_{tp}, T_{tp}	triple point (5.179 bar, 216.592 K)	Pa, K
$T_{sub}(P)$	sublimation temperature on the vapour+solid line	K
L_{sub}	latent heat of sublimation ($h_g - h_s$)	J/kg
m_g, m_l, m_s	gas / liquid / solid masses in the vessel	kg
$v_{g,l,s}, u_{g,l,s}, h_{g,l,s}$	pure-phase specific volume / internal energy / enthalpy	m ³ /kg, J/kg
α_ℓ	liquid volume fraction	–
L_v	vapour-column height above the condensate (Churchill-Chu length)	m
Ra, Nu, Pr, Gr	Rayleigh / Nusselt / Prandtl / Grashof numbers	–
h_{gw}	below-triple gas $\leftrightarrow$ wall HTC	W/m ² /K
h_{boil}	plateau wall $\leftrightarrow$ boiling-liquid HTC	W/m ² /K
h_{gs}	descent gas $\leftrightarrow$ dry-ice interphase HTC	W/m ² /K
A_g, A_{wet}, A_{gs}	gas-contact / wetted / gas-solid-interface areas	m ²
$\dot{Q}_{wg}, \dot{Q}_{wl}, \dot{Q}_{gs}, \dot{Q}$	wall-gas / wall-liquid / gas-solid / gas-liquid heat rates	W
$\dot{m}_{sub}$	sublimation mass rate (self-cooling + interphase)	kg/s

9.2 The release: input block

The CO₂ add-on is driven by a top-level **release:** block (in addition to the base vessel/initial/calculation/valve/heat_transfer sections). Its keys:

Key	Default	Meaning
type	<i>required</i>	gas (vapour-space) or liquid (bottom) release.
diameter	<i>required</i>	orifice / nozzle throat diameter [m].
discharge_coef	0.62	liquid (flashing) discharge coefficient $C_{d,liq}$.
discharge_coef_gas	0.84	gas-tail discharge coefficient $C_{d,gas}$.
liquid_nonequilibrium	0.0	HNE factor N (liquid only); 0 = HEM, 1 = frozen.
liquid_ne_pressure	false	fade N linearly toward the triple point ((4.5) in <i>Discharge Model</i>).
liquid_ne_pref	–	reference pressure for the N fade [Pa].
eos	CoolProp	thermodynamic backend (CoolProp + solid table).
solid_in_vessel	false	enable the below-triple in-vessel dry-ice model.
solid_h_gas_wall	churchill	below-triple gas-wall HTC: churchill, calc (Geankoplis), or a number.
solid_h_inner	cooper	plateau wall-liquid boiling HTC: cooper, calc (Rohsenow), or a number.
solid_h_gas_solid	3.0	descent gas $\leftrightarrow$ dry-ice interphase HTC [W/m ² /K].
solid_h_gas_liquid	0.0	plateau gas $\leftrightarrow$ liquid interphase HTC [W/m ² /K].
solid_gas_wall_frac	–	fixed override of the gas-contact wall fraction (default: from geometry).
discharge_location	0.0	liquid level [m] at which a liquid draw switches to the gas tail (heel changeover).
back_pressure	101325	downstream / choke back pressure [Pa].
atm_pressure	101325	atmospheric pressure for the downstream dry-ice flash [Pa].

Related base-model keys used by the add-on: `initial.gas_temperature` (superheated gas zone + split-wall init), `initial.wall_temperature` (uniform wall override), `calculation.non_equilibrium`: `true`, `calculation.h_gas_liquid` (calc/calc_two_sided/number), and `heat_transfer.h_outer` / `heat_transfer.h_inner`.

Note

For a **full-bore / large-diameter rupture** set `discharge_coef: 1.0` and `liquid_nonequilibrium: 0.0` (HEM); see the warning in *Hole size, residence time, and full-bore releases* on why the metastable boost must not be stacked on $C_d = 1$.

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BIBLIOGRAPHY

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